

MATH 4230 - Optimization Theory

Lecture 1

- Optimization Problem Formulation

$$(I): \min_x f(x) \quad \text{subject to } x \in K \subseteq \mathbb{R}^d \quad (P)$$

$$(II): \min_x f(x)$$

$$\text{subject to } g_i(x) \leq 0, \quad i=1,2,\dots,n$$

$$h_j(x) = 0, \quad j=1,2,\dots,m$$

Two natural questions: (i) Existence of the minimizer x^*
(ii) Property of the optimum x^* .

I. Existence of an Optimizer

Definition. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a function, and $K \subseteq \mathbb{R}^n$ be a non-empty set. We call the value $l \in [-\infty, \infty)$ the **infimum** of $f(x)$ on K if l satisfies

$$(1) \quad l \leq f(x) \quad \forall x \in K$$

$$(2) \quad \exists (x_n)_{n \geq 1} \subseteq K \text{ s.t. } \lim_{n \rightarrow \infty} f(x_n) = l$$

$$\text{Denote } l := \inf_{x \in K} f(x).$$

Corollary: (1) l always exists and is unique (l could be $-\infty$)
(2) $(x_n)_{n \geq 1}$ is called the minimizing sequence of $\inf_{x \in K} f(x)$

Definition. If $\inf_{x \in K} f(x) > -\infty$ and there exists $x^* \in K$ s.t. $f(x^*) = \inf_{x \in K} f(x)$ then we say that $f^* = f(x^*)$ is the minimum value of (P) and x^* a solution to the minimization problem (P).

Definition. A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is called **coercive** if $\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty$

Example. Let $f(x) = x^T A x + b^T x + c$ where $A \in \mathbb{R}_{++}^{n \times n}$, $b \in \mathbb{R}^n$, $c \in \mathbb{R}$. Then $f(x)$ is coercive.

Indeed, since A is positive definite, the smallest eigenvalue λ_n of A is strictly positive. Then $x^T A x \geq \lambda_n \|x\|^2$.

$$\begin{aligned} \text{Proof. } x^T A x &= (c_1 v_1 + c_2 v_2 + \dots + c_n v_n)^T A (c_1 v_1 + c_2 v_2 + \dots + c_n v_n) \quad [\text{Pick orthonormal eigenvectors } v_1, \dots, v_n] \\ &= (c_1 v_1 + c_2 v_2 + \dots + c_n v_n)^T (c_1 \lambda_1 v_1 + c_2 \lambda_2 v_2 + \dots + c_n \lambda_n v_n) \\ &= c_1^2 \lambda_1 v_1^T v_1 + c_2^2 \lambda_2 v_2^T v_2 + \dots + c_n^2 \lambda_n v_n^T v_n \\ &= \sum_{i=1}^n c_i^2 \lambda_i \quad (v_i^T v_i = 1) \\ &\geq \lambda_n \sum_{i=1}^n c_i^2 = \lambda_n \|x\|^2. \end{aligned}$$

$$b^T x \geq -(\max_{1 \leq i \leq n} |b_i|) \cdot (\sum_{i=1}^n |x_i|) \geq -M \|x\|, M \geq 0 \quad (\|x\|_p \geq \|x\|_1 \text{ for } p \geq 1)$$

$$x^T A x + b^T x + c \geq \lambda_n \|x\|^2 - M \|x\| + c \rightarrow +\infty \quad \text{when } \|x\| \rightarrow \infty$$

Proposition. Let $f_i: \mathbb{R} \rightarrow \mathbb{R}$ be bounded below and coercive for each $i=1,2,\dots,n$.
Then $f(x) = \sum_{i=1}^n f_i(x_i) \quad \forall x = (x_1, x_2, \dots, x_n) \in \mathbb{R}^n$ is also coercive.

Proof. WLOG, assume that $f_i(x_i) \geq 0$ for all $x_i \in \mathbb{R}$, $i=1,2,\dots,n$.

Then $f(x) \geq \max_{1 \leq i \leq n} f_i(x_i)$.

If $\|x\| \rightarrow +\infty$, then $\max_{1 \leq i \leq n} \|x_i\| \rightarrow +\infty$.

If $\lim_{\|x\| \rightarrow +\infty} f(x) \neq +\infty$, then $\exists (x^m)_{m \geq 1}$ s.t. $\lim_{m \rightarrow \infty} f_i(x_i^m) < +\infty \quad \forall i=1,2,\dots,n$

and $\exists i$ and a subsequence $(m_k)_{k \geq 1}$ s.t. $\|x_i^{m_k}\| \rightarrow +\infty$

Since $f_i(x_i)$ is coercive, $\lim_{m \rightarrow \infty} f_i(x_i^m) \geq \lim_{k \rightarrow \infty} f_i(x_i^{m_k}) = +\infty$

Hence is a contradiction that we have $\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty$. ■

Theorem. Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a continuous function and $K \subseteq \mathbb{R}^n$ be a non-empty subset.

① If K is compact (closed and bounded), (P) has an optimal solution $x^* \in K$

② If K is closed and $f(x)$ is coercive then (P) has a solution.

Proof. ① If K is compact, let $(x_n)_{n \geq 1}$ be a minimizing sequence of (P) i.e. $\lim_{n \rightarrow \infty} f(x_n) = \inf_{x \in K} f(x)$. Then, there exists a subsequence $(x_{n_k})_{k \geq 1}$ s.t. $\lim_{k \rightarrow \infty} x_{n_k} = x^* \in K$ so that $f(x^*) = \lim_{k \rightarrow \infty} f(x_{n_k}) = \inf_{x \in K} f(x)$. (Bounded $(x_n)_{n \geq 1}$ Bolzano-Weierstrass Theorem as K is closed as f is continuous)

② If K is closed and $f(x)$ is coercive so that $\lim_{\|x\| \rightarrow +\infty} f(x) = +\infty$

Then $\inf_{x \in K} f(x) = \inf_{x \in K_n} f(x)$ with $K_n = \{x \in K : \|x\| \leq n\}$ for n large enough.

Since K_n is closed and bounded in $\mathbb{R}^n$, it is compact.

Then, there exists $x^* \in K_n \subseteq K$ s.t. $f(x^*) = \inf_{x \in K_n} f(x) = \inf_{x \in K} f(x)$ ■

Example. $K = \{x \in \mathbb{R}^n : g(x) \leq 0, h(x) = 0\}$ is closed if g, h are continuous.

Remark. $h(x) = 0 \Leftrightarrow \begin{cases} h(x) \leq 0 \\ -h(x) \leq 0 \end{cases}$, i.e. transformation from eq. constraints to ineq. constraints

Proposition. Let $K \subseteq \mathbb{R}^n$ be a bounded and open subset and $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be continuous on $\bar{K}$ (closure of K). Then, $\exists x_0 \in \bar{K}$ s.t. $f(x_0) \leq f(x), \forall x \in \partial K = \bar{K} \setminus K$.

Then, (P) has an optimal solution $x^* \in \bar{K}$.

Proof. Since $\exists x_0 \in \bar{K}$ s.t. $f(x_0) \leq f(x) \quad \forall x \in \partial K$

we know that $\inf_{x \in \bar{K}} f(x) = \inf_{x \in K} f(x)$.

As $\bar{K}$ is bounded and closed, it is compact

Then, there exists $x^* \in \bar{K}$ s.t. $f(x^*) = \inf_{x \in \bar{K}} f(x)$

① If $x^* \in \partial K$ then $f(x_0) \leq f(x^*) = \inf_{x \in \bar{K}} f(x)$, $x_0 \in \bar{K}$ is a solution to (P).

② If $x^* \in K$ then x^* is a solution to (P). ■

II. First-order Necessary Condition of the Optimizer x^*

Theorem. If $f(x)$ is continuously differentiable, K is an open set and $x^* \in K$ is an optimal solution to (P). Then $\nabla f(x^*) = 0$. (Euler's First Order Condition)

Proof. If $\nabla f(x^*) = \begin{pmatrix} \partial_{x_1} f(x^*) \\ \vdots \\ \partial_{x_n} f(x^*) \end{pmatrix} \neq 0$, then there exists $v \in \mathbb{R}^n$ s.t. $\langle v, \nabla f(x^*) \rangle > 0$. C_1 -smooth

Define $x_\varepsilon = x^* - \varepsilon v$.

$$f(x_\varepsilon) = f(x^*) - \varepsilon \langle \nabla f(x^*), v \rangle + o(\varepsilon)$$

Then for $\varepsilon > 0$ small enough $f(x_\varepsilon) < f(x^*)$ and $x_\varepsilon \in K$ since K is open, which is a contradiction. ■

Definition (Open Set): A set S is open if $\forall x \in S, \exists r > 0 \ni B_r(x) \subset S$

Definition (Closed Set): A set S is closed if $\mathbb{R}^n \setminus S$ is open.

Definition (Supremum): Let $\emptyset \neq S \subseteq \mathbb{R}$. Suppose that S is bounded above, then $u \in \mathbb{R}$ is called a supremum ("least upper bound") of S if the following holds:

(i) u is an upper bound of S

(ii) $u \leq v$ for any upper bound v of S

equivalently, $\forall \varepsilon > 0, \exists s' \in S$ s.t. $u - \varepsilon < s'$.

① $\forall s' \in S, u \geq s'$.

($\Rightarrow$): ② For contradiction, suppose $\exists \varepsilon > 0, \forall s' \in S$ s.t. $u - \varepsilon \geq s'$.

Since $u - \varepsilon < u$, $u - \varepsilon$ is an upper bound smaller than u , a contradiction!

Another way: Fix an arbitrary $\varepsilon > 0$. Since $u - \varepsilon < u$, then by definition $u - \varepsilon$ cannot be an upper bound for S . Hence $\exists s' \in S$ s.t. $u - \varepsilon < s'$.

($\Leftarrow$): ② Let v be any upper bound of S , i.e. $v \geq s' \forall s' \in S$.

Now, let u be a supremum of S . Then, $u \geq s' \forall s' \in S$

and $\forall \varepsilon > 0, \exists s' \in S$ s.t. $u - \varepsilon < s' \leq v$

That is, $u < v + \varepsilon \forall \varepsilon > 0$. It could only be that $u \leq v$, otherwise

there is an $\varepsilon = u - v > 0$ where $u = v + \varepsilon$, as a contradiction to $u < v + \varepsilon \forall \varepsilon > 0$.

Theorem. Let (x_n) be a bounded sequence of real numbers, it has a convergent subsequence.

(The Bolzano-Weierstrass Theorem)

① Monotone Subsequence Theorem

② Monotone Convergence Theorem

(Kuhn-Tucker Theorem)

Dual feasibility

Theorem Let x^* be an optimizer of (P) . Then there exists $\vec{p} \geq \vec{0} \in \mathbb{R}^n$, $\vec{q} \in \mathbb{R}^m$.

$$\text{s.t. } \begin{cases} \textcircled{1} p_0 \nabla f(x^*) + \sum_{i=1}^n p_i \nabla g_i(x^*) + \sum_{i=1}^m q_i \nabla h_i(x^*) = 0 & \text{Stationary Condition} \\ \textcircled{2} (p_0, p_1, \dots, p_n, q_1, q_2, \dots, q_m) \neq 0 \\ \textcircled{3} \underbrace{\sum_{i=1}^n p_i g_i(x^*)}_{\geq 0 \leq 0} = 0 \Leftrightarrow p_i g_i(x^*) = 0 \quad \forall i=1, 2, \dots, n & \text{Complementary Slackness} \\ & \Leftrightarrow \text{either } p_i = 0 \text{ or } g_i(x^*) = 0 \quad \forall i=1, 2, \dots, n \end{cases}$$

Complementary Slackness

pushing to the feasible set

Proof. 1. Let $f_N(x) := f(x) + \|x - x^*\|^2 + \frac{N}{2} \left(\sum_{i=1}^n \max(0, g_i(x))^2 + \sum_{i=1}^m h_i^2(x) \right)$ [penalization function]
We have $f_N(x^*) = f(x^*)$, $f_N(x) > f(x) \quad \forall x \neq x^*$, and $f_N \in C^1$.

Remark. $x \mapsto (\max(0, x))^2 \in C^1$

2. Claim. $\exists \varepsilon_0 > 0$, s.t. $\forall \varepsilon \in (0, \varepsilon_0)$ $\exists N_\varepsilon$ s.t. $f_{N_\varepsilon}(x) > f_{N_\varepsilon}(x^*)$ for all $x \in \mathbb{R}^n$ and $\|x - x^*\| = \varepsilon$. (Proof deferred)

3. We study the optimization problem $\min_{\mathbf{p}}$

$$\min_{x \in \mathbb{R}^n} f_{N_\varepsilon}(x) \quad \text{subject to } |x - x^*| < \varepsilon.$$

There exists a solution x_2 s.t. $|x_2 - x| < \epsilon$.

By Euler's condition, $\nabla f_N(x_\varepsilon) = 0$ i.e.

By Karush-Kuhn condition, $\nabla f(x_*) = 0$ i.e.

$$\nabla f(x_*) + \lambda(x_* - x^*) + N_{\mathcal{C}}\left(\sum_{i=1}^{n'} \max(0, g_i(x_*)) \cdot \nabla g_i(x_*) + \sum_{j=1}^m h_j(x_*) \nabla h_j(x_*)\right)$$

4. Let $\rho_z = \sqrt{1 + N_z^2 \sum_{i=1}^n \max(0, g_i(x_z))^2 + N_z^2 \sum_{j=1}^m h_j(x_z)^2}$
and $\rho_0^z := \frac{1}{\rho_z} \geq 0$, $p_i^z := \frac{N_z \max(0, g_i(x_z))}{\rho_z} \geq 0$, $q_j^z := \frac{N_z h_j(x_z)}{\rho_z} \in \mathbb{R}$

Also $\| (p_0^z, p_1^z, \dots, p_n^z, q_1^z, \dots, q_m^z) \|_n = 1$

and $p_0'' \nabla f(x_2) + 2p_0''(x_2 - x^*) + \sum_{i=1}^n p_i'' \nabla g_i(x_2) + \sum_{j=1}^m q_j'' \nabla h_j(x^*) = 0$

Now, $\exists (x_i)_{i=0}^m$ s.t. $x_i \geq 0$. $(p_0, p_1, \dots, p_n, q_1, \dots, q_m) \rightarrow (p_0, p_1, \dots, p_n, q_1, \dots, q_m)$
 $x_{\varepsilon_n} \rightarrow x^*$ $\neq 0$ as the norm = 1.

$$\Rightarrow p_0 \nabla f(x^*) + \sum_{i=1}^n p_i \nabla g_i(x^*) + \sum_{j=1}^m q_j \nabla h_j(x^*) = 0$$

5. If $g_k(x^*) = 0 \rightarrow \text{okay!}$

Otherwise, $g_i(x^*) < 0$, $p_i^z = \frac{N_i \max(0, g_i(x^z))}{\rho_i} = 0$ for $z > 0$ small enough. ■

(Lecture 5)

Proof of the claim. Suppose the claim is not true, $\exists \varepsilon > 0$, s.t. $\forall N > 0$, there is $x_N \in \{z : \|z - x^*\| = \varepsilon\}$ and $f_N(x_N) \leq f_N(x^*) = f(x^*)$

The set $\partial B_\varepsilon(x^*) = \{z : \|z - x^*\| = \varepsilon\}$ is compact. Consider $\{x_N\}_{N \geq 1} \subseteq \partial B_\varepsilon(x^*)$

$$\Rightarrow \exists \text{ a subsequence } x_{N_k} \rightarrow \bar{x} \in \{z : \|z - x^*\| \leq \varepsilon\}.$$

$$\sup_{N_k} \left(C + \frac{N_k}{2} \left(\sum_{i=1}^n g_i^2(x_{N_k}) + \sum_{j=1}^m h_j^2(y_{N_k}) \right) \right) \leq \sup_{N_k} f_{N_k}(x_{N_k}) \leq f(x^*) < +\infty$$

Note. $\lim_{N \rightarrow +\infty} N \cdot l_N(x) < +\infty \Rightarrow \lim_{N \rightarrow +\infty} l_N(x) = 0$

$$\lim_{N \rightarrow +\infty} N \cdot L_N(x) < +\infty \Rightarrow \lim_{N \rightarrow +\infty} L_N(x) = 0$$

$$\sum_{i=1}^m (g_i^+(\bar{x}) + \sum_{j=1}^m h_j^2(\bar{x})) = 0 \Rightarrow \bar{x} \in K \Rightarrow \lim_{N \rightarrow +\infty} f_N(x_N) = f(\bar{x}) + \|\bar{x} - x^*\|^2 \leq f(x^*)$$

This is a contradiction to the fact that $f(\bar{x}) \geq f(x^*) \quad \forall \bar{x} \neq x^*$.

Lecture 3

Definition. We say that the constraints defined by K is qualified at $x \in K$ if $\vec{p} \geq 0_n$ and $\vec{q} \in \mathbb{R}^m$ and $\begin{cases} \sum_{i=1}^n p_i g_i(x) = 0 \\ \sum_{i=1}^n p_i \nabla g_i(x) + \sum_{j=1}^m q_j \nabla h_j(x) = 0 \end{cases}$

Then, $p_1 = p_2 = \dots = p_n = q_1 = q_2 = \dots = q_m = 0$

Theorem. Let $x^* \in K$ be a solution to (P) and further assume that the constraint K is qualified at x^* . Then, there exists $\lambda_1, \lambda_2, \dots, \lambda_n \geq 0$ and $\mu_1, \mu_2, \dots, \mu_m \in \mathbb{R}$

$$\text{s.t. } \begin{cases} \sum_{i=1}^n \lambda_i g_i(x^*) = 0 \\ \nabla f(x^*) + \sum_{i=1}^n \lambda_i \nabla g_i(x^*) + \sum_{j=1}^m \mu_j \nabla h_j(x^*) = 0 \end{cases}$$

Proof. By [KT], there exist $(p_0, p_1, \dots, p_n, q_1, q_2, \dots, q_m) \neq 0$

$$\text{s.t. } \begin{cases} \sum_{i=1}^n p_i g_i(x^*) = 0 \\ p_0 \nabla f(x^*) + \sum_{i=1}^n p_i \nabla g_i(x^*) + \sum_{j=1}^m q_j \nabla h_j(x^*) = 0 \end{cases}$$

$$\text{If } p_0 = 0, \text{ then } \begin{cases} \sum_{i=1}^n p_i g_i(x^*) = 0 \\ \sum_{i=1}^n p_i \nabla g_i(x^*) + \sum_{j=1}^m q_j \nabla h_j(x^*) = 0 \end{cases}$$

By the qualification property, $\vec{p}, \vec{q} = \vec{0}$.

This is a contradiction.

Hence $p_0 > 0$, and we can let $\lambda_i = \frac{p_i}{p_0}$, $\mu_j = \frac{q_j}{p_0}$, $i=1, 2, \dots, n$ and $j=1, 2, \dots, m$.

Example. $\min_{x \in \mathbb{R}} x$ subject $x^2 = 0$

① $f(x) = x$, $n=0$, $m=1$, $h_1(x) = x^2$.

② Check: x can only be 0 and hence $x^* = 0$

③ By **Thm**, $\exists (p_0, q_1) \neq 0$ s.t. $p_0 \nabla f(x^*) + q_1 \nabla h_1(x^*) = 0$
 $\Rightarrow p_0 = 0$

④ Conditions of **Thm** are not satisfied.

Exercise 1. $\min_{x^2+y^2=1} 2x+y$.

Solution. $l=0$ $d=2$ $m=1$. $h_1(x,y) = x^2+y^2-1$. $f(x,y) = 2x+y$.

If $(x^*, y^*) \in K$ is a solution (assume that K is qualified at (x^*, y^*)) → should be checked

Then, $\exists \mu_1 \in \mathbb{R}$ s.t. $\nabla f(x^*, y^*) + \mu_1 \nabla h_1(x^*, y^*) = 0$

$$\begin{cases} (x^*)^2 + (y^*)^2 = 1 \\ \Rightarrow \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \mu_1 \begin{pmatrix} 2x^* \\ 2y^* \end{pmatrix} = 0 \end{cases}$$

$$\Rightarrow \begin{cases} x^* = -\frac{1}{\mu_1} \\ y^* = -\frac{1}{2\mu_1} \end{cases}$$

$$\Rightarrow \mu_1 = \pm \frac{\sqrt{5}}{2} \Rightarrow \begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix} \text{ or } \begin{pmatrix} -\frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$$

By comparing $f(\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}})$ and $f(-\frac{2}{\sqrt{5}}, \frac{1}{\sqrt{5}})$ we find that $\begin{pmatrix} x^* \\ y^* \end{pmatrix} = \begin{pmatrix} \frac{2}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} \end{pmatrix}$. $f(x^*, y^*) = \sqrt{5}$.

Check of MFCA.

(i) Linear independence (automatically satisfied)

(ii) $\exists v = \begin{pmatrix} -y^* \\ x^* \end{pmatrix} \in \mathbb{R}^2$, and $\langle \begin{pmatrix} 2x^* \\ 2y^* \end{pmatrix}, \begin{pmatrix} -y^* \\ x^* \end{pmatrix} \rangle = 0$.

$$\nabla g(x,y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix}$$

Formal Justification of Thm:

$$\min_{\substack{g(x) \leq 0 \\ h(x) = 0}} f(x) = \min_{x \in \mathbb{R}^n} \sup_{\substack{\lambda \geq 0 \\ \mu \in \mathbb{R}}} f(x) + \lambda g(x) + \mu h(x)$$

Observation.

$$\sup_{\substack{\lambda \geq 0 \\ \mu \in \mathbb{R}}} f(x) + \lambda g(x) + \mu h(x) = \begin{cases} +\infty & \text{if } g(x) > 0 \text{ or } h(x) \neq 0 \\ f(x) & \text{if } x \in K. \end{cases}$$

$$\min_{x \in \mathbb{R}^n} \sup_{\substack{\lambda \geq 0 \\ \mu \in \mathbb{R}}} f(x) + \lambda g(x) + \mu h(x) \quad || \quad ?$$

$$\sup_{\substack{\lambda \geq 0 \\ \mu \in \mathbb{R}}} \min_{x \in \mathbb{R}^n} f(x) + \lambda g(x) + \mu h(x) \quad ||$$

$$\Rightarrow \min_{x \in \mathbb{R}^n} \underbrace{f(x) + \lambda^* g(x) + \mu^* h(x)}_{\text{Lagrangian}} \Rightarrow \nabla L(x^*, \lambda^*, \mu^*) = \nabla f(x^*) + \sum_{i=1}^n \lambda_i^* \nabla g_i(x^*) + \sum_{j=1}^m \mu_j^* \nabla h_j(x^*) = 0$$

Exercise 2. $\min_{x^2+y^2 \leq 1} x \cdot y$.

Existence of Solution. $f(x,y) = xy$ is continuous. $\{(x,y) : x^2+y^2 \leq 1\}$ is compact. $\Rightarrow (x^*, y^*)$ exists.

Solution. $n=1$. $f(x,y) = xy$ $g_1(x,y) = x^2+y^2-1$.

Assume that the constraint K is qualified at (x^*, y^*)

$$\text{Then } \exists \lambda_1 \geq 0, \text{ s.t. } \begin{cases} \lambda_1 g_1(x,y) = 0 \\ \nabla f(x,y) + \lambda_1 \nabla g_1(x,y) = 0 \end{cases} \Rightarrow \begin{cases} \lambda_1 (x^2+y^2-1) = 0 \\ \begin{pmatrix} y \\ x \end{pmatrix} + \lambda_1 \begin{pmatrix} 2x \\ 2y \end{pmatrix} = 0 \end{cases}$$

$$\text{Case 1. } \lambda = 0 \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow f(x,y) = 0$$

$$\text{Case 2. } \lambda > 0 \Rightarrow \begin{cases} x^2+y^2 = 1 \\ y+2\lambda x = 0 \\ x+2\lambda y = 0 \end{cases} \Rightarrow y = -2\lambda x = 4\lambda^2 y \Rightarrow \lambda = \pm \frac{1}{2} \quad (\text{note that } \lambda > 0)$$

$$\Rightarrow x = -y$$

$$\Rightarrow x = \pm \frac{\sqrt{2}}{2}, y = \mp \frac{\sqrt{2}}{2}$$

$$\Rightarrow f(x,y) = -\frac{1}{2}$$

By comparing $\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{pmatrix}, \begin{pmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{pmatrix}$ we found that both $\begin{pmatrix} \sqrt{2}/2 \\ -\sqrt{2}/2 \end{pmatrix}$ and $\begin{pmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{pmatrix}$ are optimal solutions.

Check of MFCQ.

(i) No equality constraints

$$(ii) \nabla g(x,y) = \begin{pmatrix} 2x \\ 2y \end{pmatrix} \Rightarrow \exists v = \begin{pmatrix} -2x \\ -2y \end{pmatrix} \text{ s.t. } \left\langle \begin{pmatrix} 2x \\ 2y \end{pmatrix}, \begin{pmatrix} -2x \\ -2y \end{pmatrix} \right\rangle \leq 0$$

if $(x,y) \neq 0$.

$$= -4x^2 - 4y^2$$

MFCQ holds when $(x,y) \neq 0$.

Exercise 3. $\min_{\substack{x^2+y^2=1 \\ y^2+z^2=4}} x+z$.

Solution. $d=3$, $m=2$, $h_1(x,y,z) = x^2+y^2-1$, $h_2(x,y,z) = y^2+z^2-4$, $f(x,y,z) = x+z$.

Assume that the constraints K are qualified at x^* .

$$\exists \mu_1, \mu_2 \in \mathbb{R} \text{ s.t. } \begin{cases} \nabla f(x,y,z) + \mu_1 \nabla h_1(x,y,z) + \mu_2 \nabla h_2(x,y,z) = 0 \\ x^2+y^2=1 \\ y^2+z^2=4 \end{cases}$$

$$\Rightarrow \begin{cases} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu_1 \begin{pmatrix} 2x \\ 2y \\ 0 \end{pmatrix} + \mu_2 \begin{pmatrix} 0 \\ 2y \\ 2z \end{pmatrix} = 0 \\ x^2+y^2=1 \\ y^2+z^2=4 \end{cases} \Rightarrow \begin{cases} 1+\mu_1+2\mu_2 x = 0 \\ 2y(\mu_1+\mu_2) = 0 \\ 1+2\mu_2 z = 0 \end{cases}$$

Case 1. $y=0 \Rightarrow x^2=1, z^2=4$
 $\Rightarrow \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \\ -2 \end{pmatrix}$

Case 2. $y \neq 0$
 $\begin{cases} \mu_1 + \mu_2 = 0 \\ x = -\frac{1}{2\mu_1} \\ z = \frac{1}{2\mu_2} \end{cases} \Rightarrow x = -z \Rightarrow \text{No solution.}$

We found that $f(-1,0,-2) = -3$ is the optimal solution.

$$\nabla h_1 = \begin{pmatrix} 2x \\ 2y \\ 0 \end{pmatrix}, \nabla h_2 = \begin{pmatrix} 0 \\ 2y \\ 2z \end{pmatrix} \quad \left[\text{Rank} \begin{pmatrix} 2x & 0 \\ 2y & 2y \\ 0 & 2z \end{pmatrix} = 2 \right]$$

$$\exists v = \begin{pmatrix} -yz \\ xz \\ -xz \end{pmatrix} \in \mathbb{R}^3 \text{ s.t. } v \perp \text{span}(\nabla h_1(x,y,z), \nabla h_2(x,y,z)).$$

$\Rightarrow K$ is qualified at all point $(x,y,z) \in K$.

Lecture 4

Proposition. Let $x \in K$ satisfies that:

(Karush-Kuhn-Tucker Constraint Qualification)

- ① The family of vectors $(\nabla h_1(x), \nabla h_2(x), \dots, \nabla h_m(x))$ is linearly independent
- ② There exists a vector $v \in \mathbb{R}^n$, s.t. $\langle \nabla h_j(x), v \rangle = 0$, $\forall j=1, 2, \dots, m$
 $\langle \nabla g_i(x), v \rangle < 0$, $\forall i$ s.t. $g_i(x) = 0$

Then, the constraint K is qualified at $x \in K$.

Definition. (Active Constraints) $I(x) = \{i : g_i(x) = 0\}$.

Proof. $(\nabla h_1(x), \nabla h_2(x), \dots, \nabla h_m(x))$ is linearly independent.

$$\Leftrightarrow \sum_{j=1}^m \alpha_j \nabla h_j(x) = 0 \quad \Leftrightarrow \quad \alpha_j = 0 \quad \forall j. \quad \text{————— (*)}$$

$$\Leftrightarrow \text{Rank}(\nabla h(x)) = m$$

$$\text{Furthermore, } \sum_{i=1}^n \lambda_i g_i(x) = 0 \quad \Leftrightarrow \quad \lambda_i g_i(x) = 0, \quad \forall i=1, 2, \dots, n$$

$$\Leftrightarrow \lambda_i = 0 \text{ whenever } g_i(x) < 0 \text{ and hence } i \notin I(x)$$

$$\text{Hence, } \sum_{i=1}^n \lambda_i \nabla g_i(x) + \sum_{j=1}^m \alpha_j \nabla h_j(x) = \sum_{i \in I(x)} \lambda_i \nabla g_i(x) + \sum_{j=1}^m \alpha_j \nabla h_j(x) = 0$$

$$\Rightarrow \sum_{i \in I(x)} \lambda_i \underbrace{\langle \nabla g_i(x), v \rangle}_{< 0} + \sum_{j=1}^m \alpha_j \underbrace{\langle \nabla h_j(x), v \rangle}_{= 0} = 0 \quad \text{for some vector } v \in \mathbb{R}^n$$

$$\Rightarrow \lambda_i = 0 \quad \forall i \in I(x)$$

$$\Rightarrow \lambda_i = 0 \quad \forall i \in [n].$$

$$\sum_{j=1}^m \alpha_j \nabla h_j(x) = 0 \quad \stackrel{(*)}{\Rightarrow} \quad \alpha_j = 0 \quad \forall j. \quad \blacksquare$$

Lecture 5

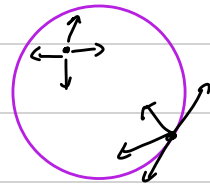
Definition (Clarke's Tangent Cone). Let $x \in K$, the tangent cone of K at the point x is defined by

$$T_K(x) := \left\{ v \in \mathbb{R}^n : \exists (s_k, v_k)_{k \geq 1} \subset \mathbb{R}_+ \times \mathbb{R}^n \text{ s.t. } s_k \searrow 0_+, \right. \\ \left. v_k \rightarrow v \text{ and } x + s_k v_k \in K, \forall k \geq 1 \right\}$$

Example. Consider $K := \{x : \|x\| \leq 1\}$

① If $x \notin \partial K$, then $T_K(x) = \mathbb{R}^2$.

② Otherwise $T_K(x) = \{\text{All vectors toward the interior of } K\}$.



Lemma. T_K is a closed cone.

Proof. ① $T_K(x)$ is indeed a cone, i.e. $v \in T_K(x) \Rightarrow \lambda v \in T_K(x) \forall \lambda \geq 0$

Case 1. $\lambda = 0$ or $v = 0$, $\lambda v = 0 \in T_K(x)$.

Case 2. $\lambda \neq 0$, $v \neq 0$

$\exists (s_k, v_k) \subseteq \mathbb{R}_+ \times \mathbb{R}^n$ satisfying the conditions of $T_K(x)$

We define $\bar{s}_k = \frac{s_k}{\lambda}$, $\bar{v}_k = \lambda v_k$

$$\Rightarrow \begin{cases} (\bar{s}_k, \bar{v}_k) = (0_+, \lambda v) \\ x + \bar{s}_k \bar{v}_k \in K \end{cases} \Rightarrow \lambda v \in T_K(x)$$

② To prove T_K is closed, we take $(v_n)_{n \geq 1} \subset T_K(x)$, $v_n \rightarrow v$.

$$\Rightarrow \exists (s_{n,k_n}, v_{n,k_n})_{n \geq 1} \text{ s.t. } s_{n,k_n} < \frac{1}{n}, |v_{n,k_n} - v_n| < \frac{1}{n}.$$

$$\text{s.t. } x + s_{n,k_n} v_{n,k_n} \in K$$

$$\text{Then, } \left. \begin{array}{l} v_{n,k_n} \rightarrow v \text{ as } n \rightarrow +\infty \\ s_{n,k_n} \rightarrow 0 \end{array} \right\} \Rightarrow v \in T_K(x).$$

Lemma. Let $K = \{x \in \mathbb{R}^n : g_i(x) \leq 0, h_j(x) = 0, \forall i=1,2,\dots,n, j=1,2,\dots,m\}$ and $x \in K$. Then $T_K(x) \subseteq \{v \in \mathbb{R}^n : \langle \nabla h_j(x), v \rangle = 0, \langle \nabla g_i(x), v \rangle \leq 0 \text{ for all } j=1,2,\dots,m, \text{ and } \forall i=1,2,\dots,n \text{ satisfying } g_i(x) = 0\}$.

Proof. Let $v \in T_K(x)$. Then $\exists (s_k, v_k) \in \mathbb{R}_+ \times \mathbb{R}^n$ s.t. $s_k \searrow 0_+$, $v_k \rightarrow v$ and $x + s_k v_k \in K$.

① $\forall i=1,2,\dots,n$ satisfying $g_i(x) = 0$, one has $g_i(x + s_k v_k) \leq 0$

$$\Rightarrow \lim_{k \rightarrow \infty} g_i(x) + s_k \langle \nabla g_i(x), v_k \rangle + s_k o(s_k) \leq 0$$

$$\Rightarrow \langle \nabla g_i(x), v \rangle \leq 0$$

② $\forall j=1,2,\dots,m$, $h_j(x + s_k v_k) = 0 \Rightarrow \lim_{k \rightarrow \infty} h_j(x) + s_k \langle \nabla h_j(x), v_k \rangle + s_k o(s_k) = 0$

$$\Rightarrow \langle \nabla h_j(x), v \rangle = 0$$

Lemma (Farkas): Let $C \in \mathbb{R}^n$, $(C_i)_{i=1, \dots, k} \subseteq \mathbb{R}^n$ $(d_j)_{j=1, 2, \dots, m} \subseteq \mathbb{R}^n$.

Assume that for all $v \in \mathbb{R}^n$

$$\left. \begin{array}{l} \langle d_j, v \rangle = 0 \quad j=1, 2, \dots, m \\ \langle C_i, v \rangle < 0 \quad i=1, 2, \dots, k \end{array} \right\} \Rightarrow \langle C, v \rangle \leq 0.$$

Then, $\exists \lambda_i \geq 0, i=1, 2, \dots, k, \mu_j \in \mathbb{R} \quad \forall j=1, 2, \dots, m$ s.t. $C = \sum_{i=1}^k \lambda_i C_i + \sum_{j=1}^m \mu_j d_j$.

Theorem (Abadie Constraint Qualification). Let x^* be a solution to (P) , suppose that $T_K(x^*) = \{v \in \mathbb{R}^n : \langle \nabla g_i(x^*), v \rangle \leq 0, \quad \forall i=1, 2, \dots, n \text{ where } g_i(x^*)=0, \\ \langle \nabla h_j(x^*), v \rangle = 0 \quad \forall j=1, 2, \dots, m\}$

Then, $\exists \lambda_1, \lambda_2, \dots, \lambda_n \geq 0, \mu_1, \mu_2, \dots, \mu_m \in \mathbb{R}$

$$\text{s.t. } \begin{cases} \sum_{i=1}^n \lambda_i g_i(x^*) = 0 \\ \nabla f(x^*) + \sum_{i=1}^n \lambda_i \nabla g_i(x^*) + \sum_{j=1}^m \mu_j \nabla h_j(x^*) = 0 \end{cases}$$

Proof. Since x^* is a minimizer,

$$\Rightarrow v \in T_K(x^*), \lim_{k \rightarrow \infty} f(x^*) \leq \lim_{k \rightarrow \infty} f(x^* + s_k v_k) = \lim_{k \rightarrow \infty} f(x^*) + s_k \langle \nabla f(x^*), v_k \rangle + o(s_k)$$

$$\Rightarrow \langle -\nabla f(x^*), v \rangle \leq 0$$

By Farkas's Lemma, $\exists \lambda_i \geq 0$ for i s.t. $g_i(x^*)=0$

$$\text{s.t. } -\nabla f(x^*) = \sum_{i: g_i(x^*)=0} \lambda_i \nabla g_i(x^*) + \sum_{j=1}^m \mu_j \nabla h_j(x^*)$$

$\Rightarrow$ For i s.t. $g_i(x^*) \neq 0$, set $\lambda_i = 0$

$\Rightarrow \lambda_i g_i(x^*) = 0 \quad \forall i=1, 2, \dots, n.$

$$\nabla f(x^*) + \sum_{i=1}^n \lambda_i \nabla g_i(x^*) + \sum_{j=1}^m \mu_j \nabla h_j(x^*) = 0.$$

Remark. If MFCQ holds for K at $x \in K$, ACQ holds at $x \in K$.

(More general)

Lecture 6.

Example 1. Let $K = \{(x, y) \in \mathbb{R}^2 : y \geq x^2, y \leq 0\} = \{(0, 0)\}$.

① $g_1(x, y) = x^2 - y, g_2(x, y) = y$

② $T_K(0) = \left\{ \begin{pmatrix} 0 \\ 0 \end{pmatrix} \right\}$

$\nabla g_1(x, y)|_{(x, y)=(0, 0)} = \begin{pmatrix} 2x \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \end{pmatrix}$

$\nabla g_2(x, y)|_{(x, y)=(0, 0)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$

$\{v \in \mathbb{R}^n : \langle v, \nabla g_i(x, y) \rangle \leq 0\} = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} : \begin{matrix} v_1 - v_2 \leq 0 \\ v_1 + v_2 \leq 0 \end{matrix} \right\} = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} : v_1 \leq 0, v_2 \in [v_1, -v_1] \right\}$.

different, ACQ does not hold.

Example 2. $K = \left\{ \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2 : y \leq x^3, y \geq 0 \right\}$

① $g_1(x, y) = y - x^3$.

$g_2(x, y) = -y$

② $\forall (x, y) \in \text{int } K, T_K(x, y) = \mathbb{R}^2, M = \mathbb{R}^2$.

③ Boundary Cases:

3.1. $x > 0, y = 0$.

$g_1(x, y) < 0, g_2(x, y) = 0$
 $T_K(x, y) = \left\{ \left\langle \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle \geq 0 \right\} = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{R}^2 : v_2 \geq 0 \right\}$
 $= M(x, y) = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \end{pmatrix} \right\} \times \{0\} = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{R}^2 : -v_2 \leq 0 \right\}$

3.2. $x > 0, y = x^3$

$T_K(x, y) = \left\{ \left\langle \begin{pmatrix} -3x^2 \\ 1 \end{pmatrix}, \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \right\rangle \leq 0 \right\} = M(x, y)$.

Following the descent direction, points towards interior of K

$-3x^2 v_1 + v_2 \leq 0$

3.3. $x = 0, y = 0$

$T_K(x, y) = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} : \begin{matrix} v_1 \geq 0 \\ -v_2 \leq 0, v_2 \leq 3xv_1 = 0 \end{matrix} \right\} = \left\{ \begin{pmatrix} v_1 \\ 0 \end{pmatrix} : v_1 \geq 0 \right\}$
 $D(x, y) = \left\{ \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} : \left\langle \begin{pmatrix} -3x^2 \\ 1 \end{pmatrix}, \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \right\rangle \leq 0, \left\langle \begin{pmatrix} 0 \\ -1 \end{pmatrix}, \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \right\rangle \leq 0 \right\}$
 $= \left\{ \begin{pmatrix} v_1 \\ 0 \end{pmatrix} : v_1 \in \mathbb{R} \right\}$.

ACQ is true for all $x \in \mathbb{R} \setminus \{(0, 0)\}$. Treat this point separately

Remark. ① For the same set K , there are different ways to formulate the constraints.

② However, $T_K(x)$ is totally geometric: it does not depend on the choice of $g(x), h(x)$'s.

Lecture 7

Example 1.
$$\begin{aligned} \min & x+y+z \\ & x^2+y^2+z^2 \leq 1 \\ & x \geq 0 \end{aligned}$$

Solution. $d=3, n=2, m=0$

$$f(x,y,z) = x+y+z$$

$$g_1(x,y,z) = x^2+y^2+z^2-1, \quad g_2(x,y,z) = -x$$

The set $K = \{(x,y,z) : g_i(x,y,z) \leq 0 \quad \forall i=1,2\}$ is compact.

Moreover, $f(x,y,z)$ is continuous, hence the optimization problem has a minimizer.

Checking the Mangasarian-Fromowitz Condition (MFCQ).

$$\nabla g_1(x,y,z) = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}, \quad \nabla g_2(x,y,z) = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}$$

To have $\langle v, \nabla g_2(x,y,z) \rangle < 0$, we must let $v_1 > 0$.

If $y=z=0$, then $\langle v, \nabla g_1(x,y,z) \rangle \geq 0$ since $x \geq 0$. The MFCQ is not satisfied.

Otherwise, we can always choose v_2, v_3 st. $v_2 \cdot 2y + v_3 \cdot 2z = -2v_1x < 0$.

The MFCQ condition is satisfied.

Case 1. $y^* = z^* = 0$. We have $\min_{\substack{x^2+y^2+z^2 \leq 1 \\ x \geq 0}} x+y+z \equiv \min_{0 \leq x \leq 1} x$

and hence it is easy to see that $x^* = 0$.

Case 2. By KKT, $\exists \lambda_1, \lambda_2 \geq 0$ st.
$$\begin{cases} \lambda_1 g_1(x^*, y^*, z^*) = 0, \lambda_2 g_2(x^*, y^*, z^*) = 0 \\ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2x^* \\ 2y^* \\ 2z^* \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \end{cases}$$

$$\Rightarrow \begin{cases} 1+2\lambda_1 x^* - \lambda_2 = 0 \\ 1+2\lambda_1 y^* = 0 \\ 1+2\lambda_1 z^* = 0 \end{cases} \Rightarrow \begin{cases} y^* = z^* = -\frac{1}{2\lambda_1} \\ x^* = \frac{\lambda_2 - 1}{2\lambda_1} \geq 0 \Rightarrow \lambda_2 \geq 1 \end{cases}$$

$$\text{Since } \lambda_1 \neq 0, g_1(x^*, y^*, z^*) = 0 \Rightarrow (x^*)^2 + (y^*)^2 + (z^*)^2 - 1 = 0$$

$$\text{Furthermore, } \lambda_2 \neq 0, g_2(x^*, y^*, z^*) = 0 \Rightarrow x^* = 0$$

$$\Rightarrow y^* = z^* = -\frac{\sqrt{2}}{2}$$

Comparing the two candidates, the optimal solution $-\sqrt{2}$ is obtained when $\begin{pmatrix} x^* \\ y^* \\ z^* \end{pmatrix} = \begin{pmatrix} 0 \\ -\frac{\sqrt{2}}{2} \\ -\frac{\sqrt{2}}{2} \end{pmatrix}$

Example 2.
$$\begin{aligned} \min & x+2y+3z \\ & x^2+y^2+z^2 = 1 \\ & x+y+z \leq 0 \end{aligned}$$

Solution. $d=3, n=m=1$. $f(x,y,z) = x+2y+3z$, $g(x,y,z) = x+y+z$, $h(x,y,z) = x^2+y^2+z^2-1$.

Checking the MFCQ condition.

$$\nabla g(x,y,z) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \quad \nabla h(x,y,z) = \begin{pmatrix} 2x \\ 2y \\ 2z \end{pmatrix}$$

① $\{\nabla h(x,y,z)\}$ is linearly independent.

② $\exists v \in \mathbb{R}^3$ st. $\langle v, \nabla h(x,y,z) \rangle = 0$ and $\langle v, \nabla g(x,y,z) \rangle \neq 0$

Note. If $\langle v, \nabla g(x,y,z) \rangle > 0$ we pick $v' = -v$ st. $\langle v', \nabla g(x,y,z) \rangle < 0$

Hence, MFCQ is satisfied.

By KKT, $\exists \lambda \geq 0, \mu \in \mathbb{R}$ st.
$$\begin{cases} \lambda(x^2+y^2+z^2-1) = 0, (x^*)^2 + (y^*)^2 + (z^*)^2 = 1 \\ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2x^* \\ 2y^* \\ 2z^* \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \mu \neq 0 \Rightarrow x^* = -\frac{1+\lambda}{\mu}, y^* = -\frac{2+\lambda}{\mu}, z^* = -\frac{3+\lambda}{\mu} \end{cases}$$

$$-\frac{\lambda(6+3\lambda)}{\mu} = 0 \Rightarrow \lambda = -2 \text{ or } 0 \Rightarrow x^* = -\frac{\sqrt{14}}{14}, y^* = -\frac{2\sqrt{14}}{14}, z^* = -\frac{3\sqrt{14}}{14} \quad \checkmark$$

$$x^* = \pm \frac{\sqrt{2}}{2}, y^* = 0, z^* = \mp \frac{\sqrt{2}}{2}$$

Lecture 8

III. Convex Optimization

① K is convex

② f (objective function) is convex.

Definition (Convex Set). A set $K \subseteq \mathbb{R}^n$ is convex if $\lambda x + (1-\lambda)y \in K$ for all $x, y \in K$ and $\lambda \in [0, 1]$.

Example of convex sets.

① $\mathbb{R}^n$

② $\mathbb{R}_+^n := \{x \in \mathbb{R}^n : x_i \geq 0, i=1, 2, \dots, n\}$.

③ Balls $B_\varepsilon(x)$

④ Intersection of convex sets

Proof. $x \in \cap K_i, y \in \cap K_i \Rightarrow x, y \in K_i \Rightarrow \lambda x + (1-\lambda)y \in K_i$

$\Rightarrow \lambda x + (1-\lambda)y \in \cap K_i$.

Definition (Point Projection onto Convex Sets) Let K be a closed, convex subset of $\mathbb{R}^n$.

For any $y \in \mathbb{R}^n$, define $\text{Proj}_K(y) := \arg\min_{x \in K} \|y - x\|$.

Remark. ① If K is not closed, $\text{Proj}_K(y)$ may not exist.

② We may consider $\arg\min_{x \in K} \|y - x\|^2$. ③ If $y \in K$, $\text{Proj}_K(y) = y$.

Proposition (Uniqueness of Projection). x^* is unique.

Proof. ① Existence: Let $f(x) := \|y - x\|, \forall x \in \mathbb{R}^n$. Clearly, $f(x)$ is coercive and K is closed, meaning that a solution exists.

② Uniqueness: Let $x_1, x_2 \in K$ and the minimizers of $\min_{x \in K} f(x)$.

Define $x_3 := \frac{x_1 + x_2}{2}$.

By Remark 2. Consider the convex function $\|y - x\|^2$

$$\begin{aligned} \text{Then, } \|y - x_3\|^2 &= \left\|y - \frac{x_1 + x_2}{2}\right\|^2 \\ &= \frac{1}{4} \|y - x_1\|^2 + \frac{1}{4} \|y - x_2\|^2 + \frac{1}{2} \langle x_1 - y, x_2 - y \rangle \end{aligned}$$

Note that by Cauchy-Schwarz inequality, $|\frac{1}{2} \langle x_1 - y, x_2 - y \rangle| \leq \frac{1}{2} \|x_1 - y\| \|x_2 - y\|$

and the equality holds iff $x_1 - y = b(x_2 - y)$ for some $b \in \mathbb{R}$.

$$\begin{aligned} \text{Hence } \|y - x_3\|^2 &< \frac{1}{4} \|y - x_1\|^2 + \frac{1}{4} \|y - x_2\|^2 + \frac{1}{2} \sqrt{\|y - x_1\|^2 \|y - x_2\|^2} \\ &= \|y - x_1\|^2, \end{aligned}$$

which is a contradiction. ■

Example 1. Consider the unit ball $K := \{x \in \mathbb{R}^n : \|x\| \leq 1\}$. Given $y \in \mathbb{R}^n, \|y\| > 1$

Find $\text{Proj}_K(y)$.

Solution. $x^* = \frac{y}{\|y\|}$, an easy observation. However, we can justify this by considering $\min_{\|x\| \leq 1} \|y - x\|^2$

Qualification. ① Automatically satisfied. ② $\nabla g(x) = 2x$. $\exists v = -x$ s.t. $v^\top \nabla g(x) < 0$ except for $g(x)$

$$\exists \lambda \geq 0 \text{ s.t. } \lambda(\|x\|^2 - 1) = 0 \text{ and } \lambda x^* - y + \lambda x^* = 0. \quad (1 + \lambda)x^* = y$$

$$(A) \lambda = 0 \Rightarrow x^* = y \text{ (which is not true)} \quad (B) \|x\| = 1 \Rightarrow x^* = \frac{y}{\|y\|}$$

Example 2. $K = \{x \in \mathbb{R}^n: a^T x = b\}$, $y \in \mathbb{R}^n$. Assuming that $a \neq \vec{0}$ and $n > 1$. Find $\text{Proj}_K(y)$.

Solution. $f(x) = \|x - y\|^2$, $h(x) = a^T x - b$.

Qualification condition: $\nabla h(x^*) = a \Rightarrow$ ① $\{\nabla h(x^*)\}$ is Linearly Independent
② $\exists v \in \mathbb{R}^n$ s.t. $\langle v, \nabla h(x^*) \rangle = 0$.

By KKT, $\exists \mu \in \mathbb{R}$ s.t.
$$\begin{cases} \nabla f(x^*) + \mu \nabla h(x^*) = \vec{0} \\ a^T x = b \end{cases}$$

$$\Rightarrow \begin{cases} 2(x^* - y) + \mu a = 0 \\ a^T x = b \end{cases}$$

$$\Rightarrow x^* = y - \frac{1}{2}\mu a$$

$$\Rightarrow a^T(y - \frac{1}{2}\mu a) = b$$

$$\Rightarrow \mu = 2(a^T y - b) / \|a\|^2$$

$$\Rightarrow x^* = y - \frac{a^T y - b}{\|a\|^2} \cdot a.$$

Lecture 9

Separation Theorem

Theorem. Let $K \subseteq \mathbb{R}^n$ be a non-empty, closed, convex set and $y \notin K$. Then, $\exists w \in \mathbb{R}^n, w \neq 0$, s.t.
 $\langle w, y \rangle < \langle w, x \rangle, \forall x \in K$. [Point-Set Separation]

Proof. Let $x^* = \text{Proj}_K(y)$ be unique. $w := x^* - y$.

$\forall x \in K$, we define $z(t) = tx + (1-t)x^* \in K, \forall t \in [0, 1]$.

$$\begin{aligned} \|y - x^*\|^2 &\leq \|y - z(t)\|^2 = \|y - tx - (1-t)x^*\|^2 = \|(y - x^*) + t(x^* - x)\|^2 \\ &= \|y - x^*\|^2 + 2t \langle y - x^*, x^* - x \rangle + t^2 \|x^* - x\|^2 \end{aligned}$$

Define $\phi(t) := \|y - z(t)\|^2$. Then, $\phi(0) = \|y - x^*\|^2 \leq \phi(t), \forall t \in [0, 1]$.

$$\phi'(t) = 2 \langle y - x^*, x^* - x \rangle + 2t \|x^* - x\|^2$$

$$\phi''(t) = 2 \langle y - x^*, x^* - x \rangle \geq 0$$

$$\Rightarrow \langle w, x \rangle - \langle w, x^* \rangle \geq 0$$

$$\Rightarrow \langle w, x \rangle \geq \langle w, x^* \rangle = \langle w, w + y \rangle = \langle w, y \rangle + \|w\|^2 > \langle w, y \rangle.$$

$$\exists w = y - x^* \neq 0, x^* = \text{Proj}_K(y) \text{ s.t. } \inf_{x \in K} w^T x = w^T x^* > w^T y$$

Corollary. Let X_1, X_2 be two closed, convex sets s.t. $X_1 \cap X_2 = \emptyset$. If X_2 is bounded, then
 $\exists w \neq 0 \in \mathbb{R}^n$ s.t. $\sup_{x \in X_2} \langle w, x \rangle < \inf_{x \in X_1} \langle w, x \rangle$. [Set-Set Separation]

Proof. Let $K := \{x_1 - x_2 : x_1 \in X_1, x_2 \in X_2\}$. K is a convex set.

Furthermore, X_2 is compact, K is closed.

Proof of Closedness of K . Let $(y_n)_{n \geq 0}$ be s.t. $y_n \rightarrow y$ and $y_n = x_n^1 - x_n^2$, $x_n^1 \in X_1, x_n^2 \in X_2$.
 Take a subsequence $x_{n_k}^2 \rightarrow x^2 \in X_2$.

Then, $x_n^1 = y_n + x_n^2 \rightarrow y + x^2 \in X_1$ and $y = x^1 - x^2 \in K$.

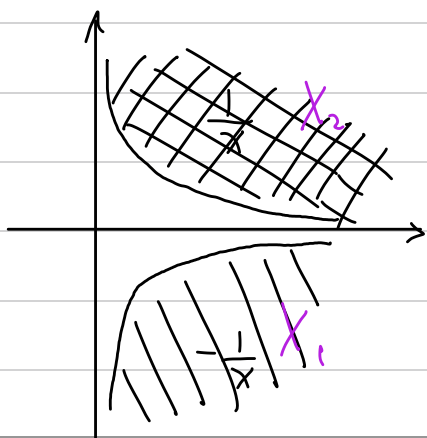
Since $X_1 \cap X_2 = \emptyset, 0 \notin K = X_1 - X_2$.

By the Point-Set Separation Theorem, $\exists w \in \mathbb{R}^n$ s.t. $\langle w, 0 \rangle > \sup_{x \in K} \langle w, x \rangle$.

$$\Rightarrow 0 > \sup_{x \in K} \langle w, x \rangle$$

$$\begin{aligned} \Rightarrow 0 &> \sup_{\substack{x_1 \in X_1 \\ x_2 \in X_2}} \langle w, x_1 - x_2 \rangle = \sup_{x_1 \in X_1, x_2 \in X_2} (\langle w, x_1 \rangle - \langle w, x_2 \rangle) = \sup_{x_1 \in X_1} \langle w, x_1 \rangle + \sup_{x_2 \in X_2} \langle w, -x_2 \rangle \\ &= \sup_{x_1 \in X_1} \langle w, x_1 \rangle - \inf_{x_2 \in X_2} \langle w, x_2 \rangle. \end{aligned}$$

Remark. What if X_2 is unbounded?



No strict inequality for separation

Can be properly separated.

Definition. A linear form $\langle w, x \rangle$ properly separates non-empty sets S and T if and only if

$$\sup_{x \in S} \langle w, x \rangle \leq \inf_{y \in T} \langle w, y \rangle$$

$$\inf_{x \in S} \langle w, x \rangle < \sup_{y \in T} \langle w, y \rangle$$

Definition. Let $X \subseteq \mathbb{R}^n$. $\text{Lin}(X)$ be the affine subspace generated by X .

The relative interior of X is the interior of X if it is considered as a subspace of $\text{Lin}(X)$.

Example. $n=2$ $X := \{(x, 0) : x \in [0, 1]\}$. $\text{Lin}(X) = \{(x, 0) : x \in \mathbb{R}\}$
 $\text{ri}(X) = \{(x, 0) : x \in (0, 1)\}$.

Example. $X_1 = \{(x, 0) : x \in [0, 1]\}$

$X_2 = \{(x, 0) : x \in [0, 2]\}$.

Take $w = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and $\langle w, x_1 \rangle = 0$, $\langle w, x_2 \rangle = 0 \quad \forall x_1 \in X_1, x_2 \in X_2$.

While $\sup_{x \in X_1} \langle w, x \rangle \leq \sup_{x \in X_2} \langle w, x \rangle$, the two sets cannot be "separated".

Hence, the strict inequality is required.

Theorem. Let X_1, X_2 be two non-empty convex sets. s.t. $\text{ri}(X_1) \cap \text{ri}(X_2) = \emptyset$. Then, X_1 and X_2 can be properly separated.

Lemma. Let S be a subset of $\mathbb{R}^n$. Then.

S is separable

$\Downarrow$
 $\exists$ a countable, dense subset of S

$\Downarrow$
 $\exists K \subseteq S$ s.t. K is countable and $\forall x \in S, \exists (x_n) \subseteq K, x_n \rightarrow x$.

Proof. Let $(r_i)_{i=1,2,\dots}$ be the set of all rational vectors so that $(r_i)_{i=1,2,\dots}$ is a countable dense subset of $\mathbb{R}^n$. Then, for each $n \geq 1$, define a set K_n as follows for each r_i , we look for a point $x \in S$ s.t. $\|x - r_i\| \leq \frac{1}{n}$ if such $x \in S$ exist, we take one and add it to K_n . Hence K_n is countable and $K_n \subseteq S$.

Let $K = \bigcup K_n \subseteq S$ and K is countable.

Let $x \in S$ be an arbitrary point. $\exists$ a sequence $r_n \rightarrow x$ where each r_n is a rational vector then for any n , $\exists m_n \geq n$ s.t. one has $x_{m_n} \in K_{m_n} \subseteq K$ and $\|x_{m_n} - r_{m_n}\| \leq \frac{1}{m_n}$.

Hence, $x_{m_n} \rightarrow x$ as $n \rightarrow +\infty$ so that K is a dense subset of S .

Lecture 10

Proposition. Let $X \subseteq \mathbb{R}^n$ be a non-empty convex set and $y \in X$.

Then, $\exists w \neq 0$ s.t. X and $\{y\}$ are properly separated by the linear form $\langle w, x \rangle$.

Proof. ① Assume that $y=0$ and $\text{Lin}(X) = \mathbb{R}^n$

Let $\{x_i\}_{i \geq 1}$ be a dense subset of X . Then for each $n \geq 1$, $\text{Conv}(\{x_i, 1 \leq i \leq n\})$ is convex, closed (and bounded).

$\exists w_n \neq 0$ s.t. $0 = w_n^T y > \max_{1 \leq i \leq n} w_n^T x_i$.

WLOG, assume that $\|w_n\| = 1$. Then, $\exists n_k$ s.t. $w_{n_k} \rightarrow w \neq 0$ by compactness

By taking the limit, we derive $0 \geq \sup_{i \geq 1} w^T x_i$

Since $\{x_i\}_{i \geq 1}$ is dense in X , $\sup_{x \in X} w^T x = \sup_{i \geq 1} w^T x_i \leq 0 = w^T y$

② Otherwise, if $\inf_{x \in X} w^T x = 0$, then $w^T x = 0 \quad \forall x \in X$

i.e. $w \perp \text{Lin}(X) = \mathbb{R}^n$, which cannot be true.

③ If $y \neq 0$, we consider $\tilde{X} = \{x - y : x \in X\}$ (shifting)

Then, if w properly separates $\{y\}$ and X , then $\{0\}$ and $\tilde{X}$ can be properly separated. We are back to Case 1.

④ If $\text{Lin}(X) \neq \mathbb{R}^n$

(a) $y \in \text{Lin}(X)$, we replace $\mathbb{R}^n$ with $\text{Lin}(X)$.

(b) $y \notin \text{Lin}(X)$, $\{y\}$ and $\text{Lin}(X)$ can be properly separated. ■

Proposition. Let X_1, X_2 be convex sets s.t. $X_1 \cap X_2 = \emptyset$. Then, X_1, X_2 can be properly separated.

Proof. Let $X = X_1 - X_2 = \{x_1 - x_2 : x_1 \in X_1, x_2 \in X_2\}$

Then, X is convex and $0 \notin X$.

By the previous proposition, $\exists w \neq 0$ s.t. $\sup_{\substack{x_1 \in X_1 \\ x_2 \in X_2}} w^T (x_1 - x_2) \leq w^T 0 = 0$
 $\inf_{\substack{x_1 \in X_1 \\ x_2 \in X_2}} w^T (x_1 - x_2) < 0$

Hence, X_1, X_2 are properly separated by $w^T x$. ■

Theorem. Let X_1, X_2 be two non-empty convex sets. s.t. $\text{ri}(X_1) \cap \text{ri}(X_2) = \emptyset$. Then, X_1 and X_2 can be properly separated.

Proof. Let $\tilde{X}_1 = \text{relint } X_1$, $\tilde{X}_2 = \text{relint } X_2$. $\tilde{X}_1$ is dense in X_1 , $\tilde{X}_2$ is dense in X_2

Then, $\tilde{X}_1 \cap \tilde{X}_2 = \emptyset$. By the previous proposition, $\tilde{X}_1, \tilde{X}_2$ can be properly separated

$$\sup_{x_1 \in \tilde{X}_1} w^T x_1 = \sup_{x_1 \in X_1} w^T x_1 \leq \inf_{x_2 \in \tilde{X}_2} w^T x_2 = \inf_{x_2 \in X_2} w^T x_2$$

and

$$\inf_{x_1 \in \tilde{X}_1} w^T x_1 = \inf_{x_1 \in X_1} w^T x_1 < \sup_{x_2 \in \tilde{X}_2} w^T x_2 = \sup_{x_2 \in X_2} w^T x_2$$

■

Lecture 11

Theorem. Let X_1, X_2 be two non-empty convex sets. If X_1, X_2 can be properly separated, then $\text{ri } X_1 \cap \text{ri } X_2 = \emptyset$

Proof. Proceed by contradiction. Suppose on the contrary that $\text{ri } X_1 \cap \text{ri } X_2 \neq \emptyset$, then $\exists \hat{x}_0 \in \text{ri } X_1 \cap \text{ri } X_2$. Hence, $\exists w \neq 0$ s.t. $\sup_{x_1 \in X_1} w^T x_1 \leq \inf_{x_2 \in X_2} w^T x_2$

$$\inf_{x_1 \in X_1} w^T x_1 < \sup_{x_2 \in X_2} w^T x_2$$

Since $w^T \hat{x}_0 \geq \inf_{x_2 \in X_2} w^T x_2 \geq \sup_{x_1 \in X_1} w^T x_1 \geq w^T \hat{x}_0$, we can only have that $\sup_{x_1 \in X_1} w^T x_1 = \inf_{x_2 \in X_2} w^T x_2 = w^T \hat{x}_0$.

If there exists $\hat{x}_1 \in X_1$ s.t. $w^T \hat{x}_1 < w^T \hat{x}_0$, then $\hat{x}_{1,\varepsilon} := \hat{x}_0 + (\hat{x}_0 - \hat{x}_1)\varepsilon \in X_1$ for $\varepsilon > 0$ small enough since $\hat{x}_0 \in \text{ri } X_1$.

$$w^T \hat{x}_{1,\varepsilon} = w^T \hat{x}_0 + \underbrace{\varepsilon w^T \hat{x}_0 - \varepsilon w^T \hat{x}_1}_{> 0} > w^T \hat{x}_0$$

This contradicts the fact that $\sup_{x \in X_1} w^T x = w^T \hat{x}_0$.

Therefore, there is no point $\hat{x}_1 \in X_1$ s.t. $w^T \hat{x}_1 < w^T \hat{x}_0 \Rightarrow w^T x_1 = w^T \hat{x}_0$ for all $x_1 \in X_1$
 $\Rightarrow \inf_{x \in X_1} w^T x = w^T \hat{x}_0$.

Similarly, $\sup_{x \in X_2} w^T x = w^T \hat{x}_0$ and $\inf_{x \in X_1} w^T x = w^T \hat{x}_0 = \sup_{x \in X_2} w^T x$, which is a contradiction to the proper separation theorem. $\text{ri } X_1 \cap \text{ri } X_2 = \emptyset$. ■

Corollary. Let $X \subseteq \mathbb{R}^n$ be a non-empty subset and $y \in X \setminus \text{ri } X$. Then, $\exists w \neq 0$ s.t. $w^T y \geq \sup_{x \in X} w^T x$ and $w^T y > \inf_{x \in X} w^T x$.

Proof. Note that $\text{ri } \{y\} = \{y\} \cap \text{ri } X = \emptyset$.

Then, $\{y\}$ and X can be properly separated.

$\exists w \neq 0$ s.t. $\sup_{x \in X} w^T x \leq w^T y$ and $\inf_{x \in X} w^T x < w^T y$

Remark. Let $y \in X \setminus \text{ri } X$ and w be the separation vector. Then, the hyperplane $\{x \in \mathbb{R}^n : w^T x = w^T y\}$ is called a supporting hyperplane of X at y .

Definition (Convex function) Let X be a non-empty convex subset of $\mathbb{R}^n$. The function $f: X \rightarrow \mathbb{R}$ is called a convex function if $f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda)f(y)$, for all $\lambda \in [0,1]$, $x, y \in X$.

Remark. f is called a strictly convex function if $f(\lambda x + (1-\lambda)y) < \lambda f(x) + (1-\lambda)f(y)$, $\forall (x, y, \lambda) \in X \times X \times [0,1]$

f is called a concave function if $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$, $\forall (x, y, \lambda) \in X \times X \times [0,1]$.

Lemma. Let $X \subseteq \mathbb{R}^n$ be convex.

① If $f: X \rightarrow \mathbb{R}$ is differentiable, then f is convex iff $f(y) \geq f(x) + \langle \nabla f(x), y-x \rangle \forall x, y \in X$

② If $f: X \rightarrow \mathbb{R}$ is twice-differentiable. Then, f is convex iff $H_f(x) \succeq 0 \quad \forall x \in X$.

Proof. ① " $\Rightarrow$ ": By convexity of f , $f(\lambda x + (1-\lambda)y) \geq \lambda f(x) + (1-\lambda)f(y)$
 $f(y) \geq \frac{f(\lambda x + (1-\lambda)y) - f(x)}{1-\lambda} - f(x) + f(x)$

$$\text{Then, } f(y) \geq \lim_{\lambda \nearrow 1} f(x) + \frac{f(\lambda x + (1-\lambda)y) - f(x)}{1-\lambda} \\ = f(x) + \langle \nabla f(x), y-x \rangle.$$

" $\Leftarrow$ ": Let $z = \lambda x + (1-\lambda)y \in X$. Then $f(x) \geq f(z) + \langle \nabla f(z), x-z \rangle$
 $f(y) \geq f(z) + \langle \nabla f(z), y-z \rangle$
 $\lambda f(x) + (1-\lambda)f(y) \geq f(z) + \langle \nabla f(z), \lambda x + (1-\lambda)y - z \rangle$

② " $\Rightarrow$ ": By convexity, if $H_f(x) < 0$ then $\exists x \in X$ and $y \in \mathbb{R}^n$ s.t. $y^T \nabla^2 f(x) y < 0$
 $\Rightarrow f(x + \varepsilon y) = f(x) + \langle \nabla f(x), \varepsilon y \rangle + \frac{1}{2} y^T \nabla^2 f(x) y + o(\varepsilon^2)$
 $< f(x) + \langle \nabla f(x), \varepsilon y \rangle$ which is a contradiction to ①.

" $\Leftarrow$ ": Let $x, y \in X$. By Taylor expansion of f ,

$$f(y) = f(x) + \langle \nabla f(x), y-x \rangle + \frac{1}{2} (y-x)^T H_f(z) (y-x) \quad \text{for some } z = \lambda x + (1-\lambda)y$$

$$\geq f(x) + \langle \nabla f(x), y-x \rangle.$$

Hence, by ①, we see that f is convex.

Example.

① $f(x) = x^2$	$f'(x) = 2x$	$f''(x) = 2$	Convex
② $f(x) = e^{ax}$	$f'(x) = ae^{ax}$	$f''(x) = a^2 e^{ax}$	Convex
③ $f(x) = -\log x$	$f'(x) = -\frac{1}{x}$	$f''(x) = \frac{1}{x^2}$	Convex [$\log(\cdot)$ is concave]
④ $f(x) = ax + b$	$f'(x) = a$	$f''(x) = 0$	Convex and Concave

⑤ $f(\vec{x}) = \frac{1}{2} \vec{x}^T A \vec{x} + b \vec{x} + c$, where $A \in \mathbb{R}_{n \times n}$ is a PSD matrix

$$f'(\vec{x}) = A\vec{x} + b \quad f''(x) = A$$

⑥ $f(x) = |x|$ Convex by triangular inequality $\|\lambda x + (1-\lambda)y\| \leq \lambda \|x\| + (1-\lambda)\|y\|$.

⑦ Sum-of-convex functions. $f(x) = \sum_{i=1}^N f_i(x)$, each $f_i(x)$ is convex.

Non-differentiable Convex Functions

Definition (Subgradients) Let X be a convex set and $f: X \rightarrow \mathbb{R}$. A vector w is a subgradient of f at point $x \in X$ if $f(x) \geq f(y) + \langle w, y - x \rangle \quad \forall y \in X$.

We denote $\partial f(x) = \{w: f(x) \geq f(y) + \langle w, y-x \rangle \forall y \in X\}$ as the subdifferentials.

Example. ① $f(x) = |x|$ $\begin{cases} -1 & \text{if } x < 0 \\ +1 & \text{if } x > 0 \\ [-1, 1] & \text{if } x = 0 \end{cases}$

Lecture 12

Definition. The epigraph of f is defined as $\text{epi } f := \{(x, t) : x \in X, t \geq f(x)\}$.

Lemma. Let X be convex, $f: X \rightarrow \mathbb{R}$. Then, f is convex if and only if $\text{epi } f$ is convex.

Proof. " $\Rightarrow$ ": Let $(x_1, t_1), (x_2, t_2) \in \text{epi } f$.

$$\text{Then } \lambda t_1 + (1-\lambda)t_2 \geq \lambda f(x_1) + (1-\lambda)f(x_2) \geq f(\lambda x_1 + (1-\lambda)x_2)$$

Hence, the point $(\lambda x_1 + (1-\lambda)x_2, \lambda t_1 + (1-\lambda)t_2) \in \text{epi } f$.

The epigraph of f , $\text{epi } f$, is convex.

" $\Leftarrow$ ": If $\text{epi } f$ is not convex, then $\exists x_1, x_2 \in X, \lambda \in (0, 1)$

$$\text{s.t. } f(\lambda x_1 + (1-\lambda)x_2) > \lambda f(x_1) + (1-\lambda)f(x_2)$$

Hence, $(\lambda x_1 + (1-\lambda)x_2, \lambda f(x_1) + (1-\lambda)f(x_2)) \notin \text{epi } f$

However, $(x_1, f(x_1)), (x_2, f(x_2)) \in \text{epi } f$

and $(\lambda x_1 + (1-\lambda)x_2, \lambda f(x_1) + (1-\lambda)f(x_2))$ is the convex combination of these two points.

We have arrived at a contradiction. ■

Proposition. Let $X \subseteq \mathbb{R}^n$ be convex, $f: X \rightarrow \mathbb{R}$.

① If $\partial f(x) \neq \emptyset, \forall x \in X$, then X is convex.

② If f is convex, then $\partial f(x) \neq \emptyset, \forall x \in \text{ri } X$.

Proof. ① Let $x, y \in X, \lambda \in (0, 1)$. Define $z := \lambda x + (1-\lambda)y \in X$

Since $\partial f(z) \neq \emptyset$, there exists $w \in \partial f(z)$ s.t.

$$f(y) \geq f(z) + \langle w, y - z \rangle$$

$$f(x) \geq f(z) + \langle w, x - z \rangle$$

$$\Rightarrow \lambda f(x) + (1-\lambda)f(y) \geq f(z) + w^T (\lambda x + (1-\lambda)y - z)$$

$$\Rightarrow f \text{ is convex.}$$

② When f is convex $\Rightarrow \text{epi } f$ is convex. Furthermore, we take $x \in \text{ri } X$

Then, $z := (x, f(x)) \in \text{epi } f$.

Since $z + (0, -\varepsilon) = (x, f(x) - \varepsilon) \notin \text{epi } f, \forall \varepsilon > 0$

Hence, z is on the relative boundary of $\text{epi } f$.

Using the Separation Theorem, we consider $X_1 = \text{epi } f \subseteq \mathbb{R}^{n+1}$ and $X_2 = \{z\}$

which are convex. Then, $\exists (w, v) \in \mathbb{R}^n \times \mathbb{R}$ s.t. $\langle (w, v), z \rangle \geq \langle (w, v), (y, t) \rangle, \forall (y, t) \in \text{epi } f$

$$\Leftrightarrow w^T x + v f(x) \geq w^T y + v t \quad \forall y \in X, t \geq f(y)$$

$$\text{① } v \leq 0, \text{ otherwise } \lim_{t \rightarrow +\infty} v t = +\infty$$

$$\text{② } v \neq 0, \text{ otherwise } w^T x \geq w^T y, \forall y \in X \Rightarrow w^T x \geq w^T (x + \varepsilon w) = w^T x + \varepsilon \|w\|^2$$

Therefore, $v < 0$, and $\frac{1}{v} w^T x + f(x) \leq \frac{1}{v} w^T y + t \quad \forall y \in X, t \geq f(y)$.

Let $t = f(y)$, then $f(y) \geq f(x) + (-\frac{w^T}{v})(y - x), \forall y \in X$.

Hence, $-\frac{w^T}{v} \in \partial f(x)$. ■

Lecture 13 Lipschitz Continuity of Convex Functions

Remark. Let $X \subseteq \mathbb{R}^n$ be convex, $f: X \rightarrow \mathbb{R}$ be a function. We extend the domain of f on $\mathbb{R}^n$ by defining $\bar{f}(x) = \begin{cases} f(x) & x \in X \\ +\infty & x \notin X \end{cases}$.

Then, f is convex on $X \iff \bar{f}$ is convex on $\mathbb{R}^n$.

Hence, we can usually consider a function $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$

$\text{dom } f = \{x \in \mathbb{R}^n : f(x) \in \mathbb{R}\}$. If $f(\cdot)$ is convex on $\mathbb{R}^n$ then $\text{dom } f$ is convex.

Lemma. Let $f(\cdot)$ be convex and $x_0 \in \text{int dom } f$. Then, $\exists \varepsilon > 0, c > 0$ s.t. $|f(x)| \leq c \quad \forall x \in B_\varepsilon(x_0) := \{x' : \|x - x'\|_2 \leq \varepsilon\}$. We say that $f(x)$ is uniformly bounded by a constant c .

Proof. Let $H_\varepsilon(x_0)$ be the hypercube around x_0 , i.e. $\{x : \|x - x_0\|_\infty \leq \varepsilon\}$.

It is clear that $B_\varepsilon(x_0) \subseteq H_\varepsilon(x_0)$.

$$\Rightarrow \sup_{x \in B_\varepsilon(x_0)} |f(x)| \leq \sup_{x \in H_\varepsilon(x_0)} |f(x)| \leq \max_{\varepsilon \in \{-1\}^n} |f(x_0 + \varepsilon \varepsilon)| =: C \quad (\text{due to the Extreme Point Theorem})$$

We know that $C < +\infty$ since $H_\varepsilon(x_0) \subseteq \text{int dom } f$. ■

Lemma. Let $f(\cdot)$ be convex, $x_0 \in \text{int dom } f$. Then, $\exists \varepsilon > 0, \tilde{C} > 0$ s.t.

$$|f(x) - f(y)| \leq C \|x - y\| \leq \tilde{C} \quad \forall x, y \in B_\varepsilon(x_0), x \neq y.$$

(Local Lipschitzness)

Proof. Since $x_0 \in \text{int dom } f$, $\exists \varepsilon, C > 0$, s.t. $B_{2\varepsilon}(x_0) \subseteq \text{int dom } f$.

and $|f(x)| \leq C \quad \forall x \in B_{2\varepsilon}(x_0)$.

Next, let z be the intersection of the line $x \rightarrow y$ and $\partial B_{2\varepsilon}(x_0)$.

Then, $\|x - y\| \leq 2\varepsilon$, $\|y - z\| \geq \varepsilon$, $\|x - z\| \geq \varepsilon$ and $y = \alpha z + (1 - \alpha)x$ where $\alpha = \frac{\|y - x\|}{\|z - x\|}$.

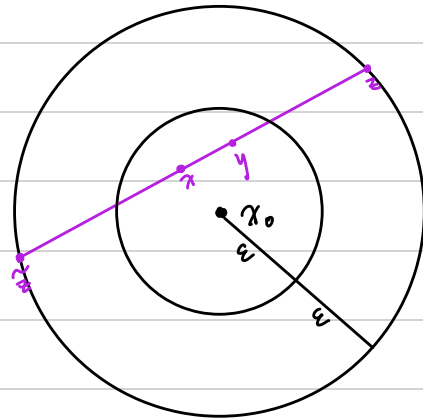
By convexity of $f(\cdot)$:

$$f(y) \leq \alpha f(z) + (1 - \alpha)f(x)$$

$$f(y) - f(x) \leq \alpha(f(z) - f(x)) = \frac{\|y - x\|}{\|z - x\|} (f(z) - f(x))$$

$$\Rightarrow \frac{f(y) - f(x)}{\|y - x\|} \leq \frac{f(z) - f(x)}{\|z - x\|} \leq \frac{2C}{\varepsilon}$$

$$\Rightarrow \frac{f(y) - f(x)}{\|x - y\|} \leq \frac{2C}{\varepsilon} = \tilde{C}$$



Theorem. Let $f(\cdot)$ be convex and $K \subseteq \text{int dom } f$ be bounded and closed.
 Then, ① f is uniformly bounded on K .
 ② f is Lipschitz on K .

Proof. K is compact. Then,

① $\{B_{\epsilon_i}(x_0) : x_0 \in K\}$ is an open cover of K

$\Rightarrow$ there exists a finite subcover $B_{\epsilon_i}(x_0)$, $i=1, \dots, m$ by compactness of K
 since $|f(x)| \leq C_i$ for all $x \in B_{\epsilon_i}(x_0)$

$\Rightarrow |f(x)| \leq \max_{1 \leq i \leq m} C_i$

② The proof is similar to the above. ■

Example. ① $f(x) = \begin{cases} x^2 & x \in (-1, +1) \\ 2 & x = \pm 1 \\ +\infty & x \notin (-1, +1) \end{cases}$

$\text{dom } f = [-1, +1]$ $\text{int dom } f = (-1, +1)$ $f(\cdot)$ is convex.
 $f(\cdot)$ is Lipschitz on $(-1, +1)$, but not Lipschitz on $[-1, +1]$.

② $f(x) = \begin{cases} \frac{1}{x} & x > 0 \\ +\infty & x \leq 0. \end{cases}$

Note that $f(\cdot)$ is convex. $\text{dom } f = (0, +\infty)$.

a. f is not uniformly bounded on $(0, +\infty)$

b. f is not uniformly Lipschitz on $(0, +\infty)$ (when $x \rightarrow 0$)

③ $f(x) = x^2$

Clearly, $f(x)$ is convex. $\text{dom } f = \mathbb{R}$.

f is not uniformly bounded and not Lipschitz on $\mathbb{R}$.

④ $f(x) = \begin{cases} -\sqrt{x} & x \geq 0 \\ +\infty & x < 0 \end{cases}$

One can verify that $-\sqrt{x}$ is convex. $\text{dom } f = [0, +\infty)$

$\text{int dom } f = (0, +\infty)$.

$K = [0, 1]$ bounded and closed. but f is not Lipschitz on K
 as $K \not\subseteq \text{int dom } f$.

Subgradients of Convex Functions

Recall that if f is convex, then $\partial f(x) \neq \emptyset \quad \forall x \in \text{relint dom } f$.

The subgradient of f at x is defined as $\{v: f(y) - f(x) \geq \langle v, y - x \rangle \quad \forall y \in \text{dom } f\}$

Lemma. ^(a) Let $f(\cdot)$ be convex, $x_0 \in \text{int dom } f$ so that $f(\cdot)$ is locally Lipschitz in the sense that

$$\exists \varepsilon, \tilde{C} \text{ s.t. } |f(x) - f(y)| \leq \tilde{C} \cdot \|x - y\| \quad \forall x, y \in B_\varepsilon(x_0).$$

Then,

$$\|v\| \leq \tilde{C} \quad \forall v \in \partial f(x) \text{ and } \forall x \in B_\varepsilon(x_0)$$

^(b) Let $v \in \partial f(x)$, $x \in \text{dom } f$, $\|v\| \leq c$. Then,

$$f(x) - f(y) \leq \langle v, x - y \rangle \leq c \|x - y\| \quad \forall y \in \text{dom } f.$$

Proof. ^(a) For $\delta > 0$ small enough, $y = x + \delta v' \in B_\varepsilon(x_0)$.

$$\Rightarrow \delta \|v\|^2 = \langle v, y - x \rangle \leq f(y) - f(x) \leq \tilde{C} \cdot \|y - x\| = \tilde{C} \cdot \delta \|v\|.$$

$$\Rightarrow \|v\| \leq \tilde{C}$$

^(b) By the definition of the subgradient and using the Cauchy-Schwarz inequality,

$$f(x) - f(y) \leq \langle v, x - y \rangle \leq \|v\| \|x - y\| \quad \forall y \in \text{dom } f.$$

$$= c \|x - y\|$$

Locally Lipschitzness $\iff$ Bounded Local Gradient

Remark. Let $f(\cdot)$ be convex and differentiable at $x \in \text{int dom } f$.

Then, $f(\lambda y + (1-\lambda)x) \leq \lambda f(y) + (1-\lambda)f(x)$.

$$\Rightarrow f(y) \geq \frac{f((1-\lambda)x + \lambda y) - (1-\lambda)f(x)}{\lambda} = f(x) + \frac{1}{\lambda} (f((1-\lambda)x + \lambda y) - f(x))$$

Let $\lambda \searrow 0$, we have

$$f(y) \geq f(x) + \langle \nabla f(x), y - x \rangle.$$

Properties of Subgradients

^(a) Scaling: $\partial(af) = a \partial f$ for $a > 0$

^(b) Sum: $\partial(f+g) = \partial f + \partial g$

Theorem. Let f_1, f_2 be convex functions and $\text{ri dom } f_1 \cap \text{ri dom } f_2 \neq \emptyset$

Then, $\partial(f_1 + f_2)(x) = \partial f_1(x) + \partial f_2(x)$ for all $x \in \text{dom } f_1 \cap \text{dom } f_2$.

Proof Sketch (1-Dimensional).

Claim. $\partial f(x) = [\partial_- f(x), \partial_+ f(x)]$ where $\partial_- f(x) = \lim_{y \nearrow x} \frac{f(y) - f(x)}{y - x}$, $\partial_+ f(x) = \lim_{y \searrow x} \frac{f(y) - f(x)}{y - x}$.

In fact, $\partial_- f(x)$, $\partial_+ f(x)$ must exist, since by convexity, $y_2 > y_1 > x \Rightarrow \frac{f(y_2) - f(x)}{y_2 - x} \geq \frac{f(y_1) - f(x)}{y_1 - x} \geq \partial_+ f(x)$

$f(y_2) - f(x) \geq \partial_+ f(x) \cdot (y_2 - x) \quad \forall y_2 > x \Rightarrow \partial_+ f(x) + \varepsilon \notin \partial f(x) \quad \forall \varepsilon > 0$ [Similar for the other case]

Secondly, $\partial(f_1 + f_2)(x) = [\partial f_1(x) + \partial f_2(x), \partial f_1(x) + \partial f_2(x)]$.

Lecture 14 Optimality Condition of Convex Functions

Proposition. Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex. Then, $\forall x \in \text{dom } f$, the following statements are equivalent:

- (a) x is a local minimum of $f(\cdot)$;
- (b) x is a global minimum of f ;
- (c) $0 \in \partial f(x)$.

Proof. (c) $\Rightarrow$ (b): If $0 \in \partial f(x)$, then $f(y) \geq f(x) + \langle 0, y-x \rangle \quad \forall y \in \mathbb{R}^n$.

Hence x is a local minimum

(b) $\Rightarrow$ (a):

(c) $\Rightarrow$ (a): Trivial.

(a) $\Rightarrow$ (c): If x is a local minimum of f , then $f(x) \leq f(z) \quad \forall z \in B_\varepsilon(x)$

Let $y \in \mathbb{R}^n$ for $\lambda > 0$ small enough, $x + \lambda(y-x) \in B_\varepsilon(x)$. Then,

$$f(x) \leq f(x + \lambda(y-x)) = f(\lambda y + (1-\lambda)x) \leq \lambda f(y) + (1-\lambda)f(x).$$

$$\lambda f(y) \geq \lambda f(x) + \langle 0, y-x \rangle = \lambda f(x) \quad \forall y$$

$$\Rightarrow 0 \in \partial f(x).$$

Remark. Let $X \subseteq \mathbb{R}^n$ be a convex set and define $I_X(x) = \begin{cases} 0, & x \in X \\ +\infty, & x \notin X \end{cases}$ which is a convex function.

The subgradient of I_X , $\partial I_X(x)$, is $\{v \in \mathbb{R}^n: \langle v, y-x \rangle \leq 0, \forall y \in X\}$ when $x \in X$.

Proposition. Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a convex function, $X = \text{dom } f$. Then

$x^* \in X$ be a solution to $\min_{x \in \mathbb{R}^n} f(x) = \min_{x \in X} f(x) \Leftrightarrow \exists v^* \in \partial f(x^*)$ s.t. $\langle v^*, y-x^* \rangle \geq 0, \forall y \in X$.

Proof. x^* is a solution to $\min_{x \in X} f(x) \Leftrightarrow x^*$ is a solution to $\min_{x \in \mathbb{R}^n} f(x) + I_X(x)$.

$$\Leftrightarrow 0 \in \partial(f + I_X)(x^*) = \partial f(x^*) + \partial I_X(x^*)$$

$$\Leftrightarrow \exists v^* \in \mathbb{R}^n \text{ s.t. } v^* \in \partial f(x^*), -v^* \in \partial I_X(x^*)$$

$$\Leftrightarrow \exists v^* \in \partial f(x^*) \text{ and } \langle -v^*, y-x^* \rangle \leq 0 \quad \forall y \in X$$

$$\Leftrightarrow \exists v^* \in \partial f(x^*) \text{ s.t. } \langle v^*, y-x^* \rangle \geq 0, \forall y \in X$$

Lecture 15 Lagrange Duality for Convex Optimization Problems

Consider the problem (P):

$$\min_{x \in \mathbb{R}^n} f(x) \quad \text{s.t.} \quad \begin{aligned} g_i(x) &\leq 0, \quad i=1,2,\dots,n \\ h_j(x) &= 0, \quad j=1,2,\dots,m \end{aligned}$$

where $f, g_i \in \mathbb{R}^n \rightarrow \mathbb{R}$ are convex functions, $h_j: \mathbb{R}^n \rightarrow \mathbb{R}$ are affine functions, i.e. $h_j(x) = a_j^T x + b_j$.

Definition (Lagrangian) The Lagrangian function of (P) is defined by:

$$L(x, \lambda, \mu) := f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x)$$

for $\lambda_1, \lambda_2, \dots, \lambda_n \in \mathbb{R}_{++}$, $\mu_1, \mu_2, \dots, \mu_m \in \mathbb{R}$

Remark. The primal Lagrangian problem

$$\min_{x \in \mathbb{R}^n} \sup_{\substack{\lambda_i \in \mathbb{R}_{++} \\ \mu_j \in \mathbb{R}}} L(x, \lambda, \mu)$$

is equivalent to (P), since

$$\sup_{\substack{\lambda_i \in \mathbb{R}_{++} \\ \mu_j \in \mathbb{R}}} L(x, \lambda, \mu) = \begin{cases} f(x), & \text{if } g_i(x) \leq 0, h_j(x) = 0 \quad \forall i=1,2,\dots,n, \text{ and } j=1,2,\dots,m. \\ +\infty, & \text{otherwise} \end{cases}$$

Definition (Dual Optimization Problem)

$$\max_{\substack{\lambda_i \in \mathbb{R}_{++} \\ \mu_j \in \mathbb{R}}} d(\lambda, \mu) \quad \text{where } d(\lambda, \mu) := \inf_{x \in \mathbb{R}^n} L(x, \lambda, \mu)$$

Proposition (Weak Duality) $\inf_{a \in A} \sup_{b \in B} L(a, b) \geq \sup_{b \in B} \inf_{a \in A} L(a, b)$

Proof. $\forall a \in A$,

$$\begin{aligned} \inf_{a \in A} L(a, b) &\leq L(a, b) \quad \forall a, b \\ \Rightarrow \underbrace{\sup_{b \in B} \inf_{a \in A} L(a, b)}_c &\leq \underbrace{\sup_{b \in B} L(a, b)}_{f(a)}, \quad \forall a. \end{aligned}$$

$$\Rightarrow c \leq \inf_{a \in A} f(a)$$

Theorem (Strong Duality) Let f, g_i be convex and differentiable, $h_j(x)$ be affine.

Assume that $f^* > -\infty$ is bounded below, and there exists $\bar{x} \in \mathbb{R}^n$ st. $g_i(\bar{x}) < 0 \quad \forall i, h_j(\bar{x}) = 0 \quad \forall j$.

Then, $\exists (\lambda^*, \mu^*)$ st. $d(\lambda^*, \mu^*) = \max_{\substack{\lambda_i \in \mathbb{R}_{++} \\ \mu_j \in \mathbb{R}}} d(\lambda, \mu) = f^*$

To further simplify the problem, consider the constraints formulated by (I) and (II)

$$(I): f(x) \leq c$$

$$g_i(x) \leq 0, \quad i=1,2,\dots,n$$

$$h_j(x) = 0, \quad j=1,2,\dots,m$$

$$(II): \lambda_i \geq 0, \quad i=1,2,\dots,n$$

$$\inf_x (f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x)) \geq c$$

Proposition. If there exists $\lambda_i, i=1,2,\dots,n$ satisfying (II)

then there is no $x \in \mathbb{R}^n$ satisfying (I).

Proof. Assume that (II) is solvable, then $\exists \lambda_i \geq 0$ s.t. $\inf_{x \in \mathbb{R}^n} (f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x)) \geq c$
i.e. $f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x) \geq c \quad \forall x \in \mathbb{R}^n$

Next, if $x \in \mathbb{R}^n$ satisfies $g_i(x) \leq 0, h_j(x) = 0$, then $f(x) \geq c - \underbrace{\sum_{i=1}^n \lambda_i g_i(x)}_{\geq 0} - \underbrace{\sum_{j=1}^m \mu_j h_j(x)}_{\leq 0} \geq c$
meaning that (I) is insolvable. ■

Proposition. If (I) is insolvable and $\begin{cases} g_i(\bar{x}) < 0 & i=1,2,\dots,n \\ h_j(\bar{x}) = 0 & j=1,2,\dots,m \end{cases}$ is solvable

Then, (II) is solvable.

Proof. (I) is insolvable $\Rightarrow \forall x \in \mathbb{R}^d$ s.t. $g_i(x) \leq 0, \forall i=1,2,\dots,n, h_j(x) = 0 \quad \forall j=1,2,\dots,m$,
we have $f(x) \geq c$

————— cont. on Lec 16. ■

Lecture 16. Definition (Slater's Condition) $\exists x$ s.t. $g_i(x) < 0, h_j(x) = 0 \quad \forall i=1,2,\dots,n, j=1,2,\dots,m$.

(I) $f(x) < C$ $g_i(x) \leq 0, i=1,2,\dots,n$ $h_j(x) = 0, j=1,2,\dots,m$	(II) $d(\lambda, \mu) \geq C$ $\lambda_i \geq 0, i=1,2,\dots,n$ $\mu_j \in \mathbb{R}, j=1,2,\dots,m$
---	---

Proposition (Reformulation). If (I) is insolvable, and Slater condition holds, then (II) is solvable.

Proof. Let $S := \{(u_0, u, v) : u_0 \in \mathbb{R}, u \in \mathbb{R}^n, v \in \mathbb{R}^m, u_0 < 0, u_i \leq 0, v_j = 0\}$.

$T := \{(u_0, u, v) : \exists x \in \mathbb{R}^d, \text{ s.t. } f(x) \leq u_0, g_i(x) \leq u_i (i=1,2,\dots,n), h_j(x) = v_j (j=1,2,\dots,m)\}$

S is non-empty and convex. T is non-empty and convex. $S \cap T = \emptyset$. Otherwise, (I) is solvable.

By Separation Theorem, $\exists w \in \mathbb{R}^{1+n+m}$ s.t.

$$\sup_{\substack{u_0 < 0 \\ u_i \leq 0 \\ v_j = 0}} (w_0 u_0 + \sum_{i=1}^n w_i u_i + \sum_{j=1}^m w_{n+j} v_j) = \sup_{(u_0, u, v) \in S} \langle w, (u_0, u, v) \rangle \leq \inf_{(u_0, u, v) \in T} \langle w, (u_0, u, v) \rangle = \inf_{\substack{u_0 \geq f(x) \\ u_i \geq g_i(x) \\ v_j = h_j(x)}} (w_0 u_0 + \sum_{i=1}^n w_i u_i + \sum_{j=1}^m w_{n+j} v_j) < +\infty$$

Since $\sup_{\substack{u_0 < 0 \\ u_i \leq 0}} (w_0 u_0 + \sum_{i=1}^n w_i u_i) < +\infty$, we have $\begin{cases} w_0 \geq 0 \\ w_i \geq 0, i=1,2,\dots,n \end{cases}$ and $\sup_{\substack{u_0 < 0 \\ u_i \leq 0}} (w_0 u_0 + \sum_{i=1}^n w_i u_i) = w_0 C$.

On the other hand, $\inf_{\substack{x \in \mathbb{R}^d \\ u_i \geq g_i(x) \\ v_j = h_j(x)}} (w_0 u_0 + \sum_{i=1}^n w_i u_i + \sum_{j=1}^m w_{n+j} v_j) = \inf_{x \in \mathbb{R}^d} (w_0 f(x) + \sum_{i=1}^n w_i g_i(x) + \sum_{j=1}^m w_{n+j} h_j(x)) \geq w_0 C$.

① If $w_0 = 0, w_0 C = 0 \leq \inf_{x \in \mathbb{R}^d} (w_0 f(x) + \sum_{i=1}^n w_i g_i(x) + \sum_{j=1}^m w_{n+j} h_j(x)) < 0$, which is a contradiction.

② Otherwise, $w_0 > 0$ and $\inf_{x \in \mathbb{R}^d} (f(x) + \sum_{i=1}^n \frac{w_i}{w_0} g_i(x) + \sum_{j=1}^m \frac{w_{n+j}}{w_0} h_j(x)) \geq C$,

$\Leftrightarrow d(\frac{w_i}{w_0}, \frac{w_{n+j}}{w_0}) \geq C$, meaning that (II) is solvable. ■

Saddle Points

Theorem. (i) Let $x^* \in \mathbb{R}^d, \lambda^* \in \mathbb{R}_+^n, \mu^* \in \mathbb{R}^m$ be such that $L(x, \lambda, \mu) \geq L(x^*, \lambda^*, \mu^*) \geq L(x^*, \lambda, \mu)$

$\forall x \in \mathbb{R}^d, \forall (\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m$. Then, x^* is a solution to $(P) \equiv \inf_{x \in \mathbb{R}^d} f(x)$ subject to $g_i(x) \leq 0, h_j(x) = 0$.

(ii) Let x^* be a solution to (P) and (P) satisfies Slater's condition. Then, $\exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$

such that $L(x, \lambda^*, \mu^*) \geq L(x^*, \mu^*, \lambda^*) \geq L(x^*, \mu, \lambda)$, $\forall x \in \mathbb{R}^d, (\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m$.

Proof. (i) $L(x^*, \lambda^*, \mu^*) \geq L(x^*, \lambda, \mu) \quad \forall (\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m$

$$\Rightarrow L(x^*, \lambda^*, \mu^*) \geq \sup_{(\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m} f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x) = \begin{cases} f(x^*), & x \in K \\ +\infty, & \text{otherwise} \end{cases}$$

Since $L(x, \lambda^*, \mu^*)$ is finite and bounds $L(x^*, \lambda^*, \mu^*)$, $x^* \in K$ and $g_i(x^*) \leq 0$ and $h_j(x^*) = 0$.

$$\Rightarrow f(x^*) = L(x^*, \lambda^*, \mu^*) \leq \inf_{x \in \mathbb{R}^d} L(x, \lambda^*, \mu^*) = d(\lambda^*, \mu^*)$$

$$\Rightarrow f(x^*) \leq \sup_{(\lambda, \mu)} d(\lambda, \mu) = D \leq P. \quad (\text{by weak duality}) \quad x^* \text{ is a solution to } (P).$$

(ii) By Duality, $\exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$ s.t. $D = P = d(\lambda^*, \mu^*) = \inf_{x \in \mathbb{R}^d} L(x, \lambda^*, \mu^*)$

$$\Rightarrow f(x^*) = P \leq \inf_{x \in \mathbb{R}^d} L(x, \lambda^*, \mu^*) \leq f(x^*) + \sum_{i=1}^n \lambda_i^* g_i(x^*) + \sum_{j=1}^m \mu_j^* h_j(x^*)$$

Since $x^* \in K$, we know that $g_i(x^*) < 0, h_j(x^*) = 0$, i.e. $\sum_{i=1}^n \lambda_i^* g_i(x^*) < 0, \sum_{j=1}^m \mu_j^* h_j(x^*) = 0$

Then, $\lambda_i g_i(x^*) = 0 \quad \forall i=1,2,\dots,n, \quad L(x, \lambda^*, \mu^*) \geq L(x^*, \lambda^*, \mu^*) \quad \forall x \in \mathbb{R}^d$

$$\begin{aligned} &= f(x^*) \\ &\geq f(x^*) + \sum_{i=1}^n \lambda_i^* g_i(x^*) + \sum_{j=1}^m \mu_j^* h_j(x^*) \\ &= L(x^*, \lambda^*, \mu^*), \quad \forall (\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m. \end{aligned}$$

Lecture 17

Let f, g_i be convex, h_j be affine.

Consider the two optimization problems:

$$P := \inf_{x \in K} f(x), \quad K := \{x \in \mathbb{R}^d : g_i(x) \leq 0, \forall i=1,2,\dots,n \text{ and } h_j(x)=0 \forall j=1,2,\dots,m\}$$

$$D := \sup_{(\lambda, \mu) \in \mathbb{R}_+^n \times \mathbb{R}^m} d(\lambda, \mu)$$

$$\text{where } d(\lambda, \mu) = \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda, \mu), \text{ and } \mathcal{L}(x, \lambda, \mu) := f(x) + \sum_{i=1}^n \lambda_i g_i(x) + \sum_{j=1}^m \mu_j h_j(x).$$

Theorem. Under Slater's condition, $^{(1)} P = D$ $^{(2)} \exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$ s.t. $P = D = d(\lambda^*, \mu^*)$

Proof. $^{(1)}$ By weak duality, $P \geq D$.

$^{(2)}$ Set $C = P$ in (I), (II) so that (I) is insolvable.

By Proposition, (II) is solvable. $\exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$ s.t. $D \geq d(\lambda^*, \mu^*) \geq C = P$

$$\Rightarrow D = d(\lambda^*, \mu^*) = P$$

Proposition. Let $g_i(x)$ be convex and $h_j(x) = A_j^T x + b_j$. Assume that $\{A_1, A_2, \dots, A_n\}$ is linearly independent and the Slater's condition holds. Then, qualification condition holds at all $x \in K$.

Proof. $^{(1)}$ $\{h_1(x), h_2(x), \dots, h_m(x)\}$ is linearly independent.

$^{(2)}$ $\forall x \neq \hat{x}$. Let $v := (\hat{x} - x)$. $0 = h_j(\hat{x}) - h_j(x) = A_j^T (\hat{x} - x) = -\langle \nabla h_j(x), v \rangle$.

Further, when $g_i(x) = 0$, $\langle \nabla g_i(x), v \rangle = \langle \nabla g_i(x), \hat{x} - x \rangle \leq g_i(\hat{x}) - g_i(x) = g_i(\hat{x}) < 0$

$$g_i(x) \leq 0, \quad h_j(x) = 0$$

$$g_i(\hat{x}) < 0, \quad h_j(\hat{x}) = 0$$

Lecture 18 Corollary. Let $x^* \in K$ be an optimal solution to $\mathcal{P}$ and it satisfies the qualification condition. Then, $\exists \lambda^* \in \mathbb{R}_+^n, \mu^* \in \mathbb{R}^m$ s.t. $\sum_{i=1}^n \lambda_i^* g_i(x^*) = 0$ and $\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0$

Theorem. Suppose that the constraint K satisfies the qualification condition and $x^* \in K$. Then, x^* is an optimal solution to $(\mathcal{P})$ if and only if $\exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$ s.t. $\sum_{i=1}^n \lambda_i^* g_i(x^*) = 0$ and $\nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0$

Proof. It remains to prove the $(\Leftarrow)$ direction. Since $x \mapsto \mathcal{L}(x, \lambda, \mu)$ is convex, we have $\mathcal{L}(x^*, \lambda^*, \mu^*) = \min_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda^*, \mu^*) = d(\lambda^*, \mu^*)$. Hence,

$$\forall y \in \mathbb{R}^d, \mathcal{L}(y, \lambda^*, \mu^*) \geq \mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_{i=1}^n \lambda_i^* g_i(x^*) + \sum_{j=1}^m \mu_j^* h_j(x^*) = f(x^*)$$

$$\text{Moreover, } \forall y \in K, \mathcal{L}(y, \lambda^*, \mu^*) = f(y) + \sum_{i=1}^n \lambda_i^* g_i(y) \leq f(y) \Rightarrow f(x^*) \leq \mathcal{L}(y, \lambda^*, \mu^*) \leq f(y)$$

Hence, x^* is an optimal solution to $(\mathcal{P})$. ■

Theorem (Duality by KKT) Assume that K satisfies the qualification condition and $(\mathcal{P})$ has a solution. Then, $P = D$ and $\exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m$ s.t. $D = d(\lambda^*, \mu^*)$. Moreover, $\exists x^*$ s.t. $\mathcal{L}(x^*, \lambda^*, \mu^*) = d(\lambda^*, \mu^*)$ and x^* is a solution to $(\mathcal{P})$. (Remark. $(\mathcal{P})$ is a convex optimization problem.)

Proof. By weak duality, $P \geq D$ is always true.

$$\begin{aligned} \text{Let } x^* \text{ be a solution to } (\mathcal{P}), \exists (\lambda^*, \mu^*) \in \mathbb{R}_+^n \times \mathbb{R}^m \text{ s.t. } \sum_{i=1}^n \lambda_i^* g_i(x^*) = 0, \nabla_x \mathcal{L}(x^*, \lambda^*, \mu^*) = 0 \\ D = \sup_{\lambda \in \mathbb{R}_+^n, \mu \in \mathbb{R}^m} d(\lambda, \mu) \geq d(\lambda^*, \mu^*) \equiv \inf_{x \in \mathbb{R}^d} \mathcal{L}(x, \lambda^*, \mu^*) = \mathcal{L}(x^*, \lambda^*, \mu^*) = f(x^*) + \sum_{i=1}^n \lambda_i^* g_i(x^*) + \sum_{j=1}^m \mu_j^* h_j(x^*) \\ = \inf_{x \in K} f(x) = P = d(\lambda^*, \mu^*) = \mathcal{L}(x^*, \lambda^*, \mu^*) \end{aligned}$$

The fact that $D = d(\lambda^*, \mu^*)$ implies that (λ^*, μ^*) is a dual optimizer

On the other hand, x^* is optimizer in $d(\lambda^*, \mu^*)$ and x^* is optimizer for $(\mathcal{P})$ ■

Exercise 1. $\min_{\substack{-\log(x) - y \leq 0 \\ y \geq 1}} x + \frac{1}{2} y^2$

Solution. $f(x, y) = x + \frac{1}{2} y^2$

$$g_1(x, y) = -\log x - y$$

$$g_2(x, y) = -y + 1$$

① We verify that $f(x, y)$, $g_1(x, y)$ and $g_2(x, y)$ are all convex.

② Existence of solution:

$$K := \{(x, y) : y \geq 1, x \geq e^{-y}\}$$

(a) f is coercive. (b) K is closed.

$$\textcircled{3} \mathcal{L}(x, y, \lambda_1, \lambda_2) = x + \frac{1}{2} y^2 + \lambda_1 (-\log x - y) + \lambda_2 (-y + 1)$$

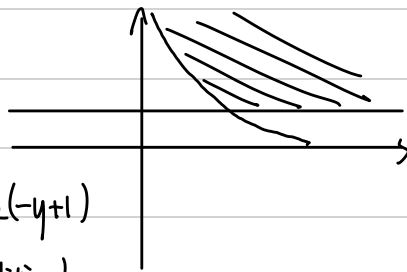
$$\nabla_{(x, y)} \mathcal{L}(x, y, \lambda_1, \lambda_2) = \begin{pmatrix} 1 - \frac{\lambda_1}{x} \\ y - \lambda_1 - \lambda_2 \end{pmatrix} = 0 \quad (\text{Euler's condition})$$

$$x = \lambda_1, y = \lambda_1 + \lambda_2 \quad \mathcal{L}(x, y, \lambda_1, \lambda_2) = \lambda_1 + \frac{1}{2} (\lambda_1 + \lambda_2)^2 - \lambda_1 \log \lambda_1 - \lambda_1 (\lambda_1 + \lambda_2) + \lambda_2 (\lambda_1 - \lambda_1 + 1)$$

$$= \lambda_1 - \frac{1}{2} \lambda_1^2 + \frac{1}{2} \lambda_2^2 - \lambda_1 \lambda_2 - \lambda_1 \log \lambda_1 + \lambda_2$$

$$\nabla_{(\lambda_1, \lambda_2)} \mathcal{L}(x, y, \lambda_1, \lambda_2) = \begin{pmatrix} 1 - \log \lambda_1 - 1 - (\lambda_1 + \lambda_2) \\ -(\lambda_1 + \lambda_2) + 1 \end{pmatrix} = 0 \Rightarrow \lambda_1^* = e^{-(\lambda_1^* + \lambda_2^*)} = e^{-1}, \lambda_2^* = 1 - e^{-1}$$

$$f(x^*, y^*) = e^{-1} + \frac{1}{2}$$



Exercise 2. Let $a_i > 0$, $i=1, 2, \dots, n$. $K := \{x \in \mathbb{R}^d : \sum_{i=1}^n \frac{x_i^2}{a_i^2} \leq 1\}$, $\min_{x \in K} \|x-u\|^2$ for given $u \in \mathbb{R}^d$.

Solution. $\|x-u\|^2 \geq 0$. Hence, if $u \in K$ we may let $x=u \in K$ to obtain the minimum.

$$f(x) = \|x-u\|^2, g(x) = \sum_{i=1}^n \frac{x_i^2}{a_i^2} - 1. \quad \mathcal{L}(x, \lambda) = \|x-u\|^2 + \lambda \left(\sum_{i=1}^n \frac{x_i^2}{a_i^2} - 1 \right).$$

$$[\nabla_x \mathcal{L}(x, \lambda)]_i = 2(x_i - u_i) + \lambda(2x_i/a_i^2), \quad x_i' = \frac{u_i}{1 + \lambda/a_i^2}$$

$$\mathcal{L}(x', \lambda) = \sum_{i=1}^n \left(1 - \frac{1}{1 + \lambda/a_i^2}\right)^2 u_i^2 + \lambda \left(\sum_{i=1}^n \frac{u_i^2}{(1 + \lambda/a_i^2)^2} a_i^2 - 1 \right)$$

$$d(\lambda) = -1 + \lambda \sum_{i=1}^n \frac{u_i^2}{a_i^2 + \lambda}$$

$$\sup_{\lambda \geq 0} d(\lambda) \Rightarrow -1 + \sum_{i=1}^n \frac{u_i^2}{a_i^2 + \lambda} - \lambda \sum_{i=1}^n \frac{u_i^2}{(a_i^2 + \lambda)^2} = 0$$

$$1 = \sum_{i=1}^n \frac{u_i^2 (a_i^2 - \lambda + \lambda)}{(a_i^2 + \lambda)^2} \Rightarrow x_i^* = \frac{u_i a_i}{a_i^2 + \lambda^*} = a_i \frac{u_i a_i}{(a_i^2 + \lambda^*)}$$

Lecture 19

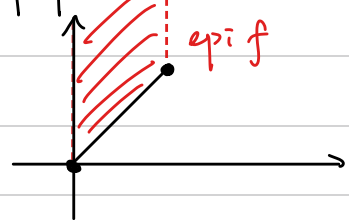
Legendre-Fenchel Conjugate

Recall that for a convex set $X \subseteq \mathbb{R}^d$ and a convex $f: X \rightarrow \mathbb{R}$ we can extend $\bar{f}(x) = \begin{cases} f(x), & \text{if } x \in X \\ \infty, & \text{if } x \notin X \end{cases}$ and $\bar{f}(x)$ is convex.

This enables us to consider the general function $\bar{f}: \mathbb{R}^d \rightarrow \mathbb{R} \cup \{\infty\}$ with $\text{dom } f = \{x: f(x) \in \mathbb{R}\}$

Also, $\bar{f}$ is locally-Lipschitz on $\text{relint dom } f$, and note that $\bar{f}$ is not continuous on the boundary of $\text{dom } f$.

Definition (Semi-continuity) Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$. We define $\text{epi } f := \{(x, t): x \in \mathbb{R}^d, t \geq f(x)\} \subseteq \mathbb{R}^{d+1}$ and f is lower semi-continuous if $\liminf_{x \rightarrow \bar{x}} f(x) \geq f(\bar{x})$ and a convex f is proper if $f(x) \in \mathbb{R} \cup \{+\infty\}$ $\forall x \in \mathbb{R}^d$ and $\exists \hat{x} \in \mathbb{R}^d$ s.t. $f(\hat{x}) \in \mathbb{R}$.



Example. (Lower-Semicontinuous functions)

① Let $A = [0, 1]$ $f(x) = x \mathbb{1}_{[x \in [0, 1]]} + \infty \mathbb{1}_{[x \notin [0, 1]]}$ is l.s.c.

$f(x)$ is continuous on $(0, 1)$. $\liminf_{x \rightarrow 0^-} f(x) = \infty \geq f(0)$, $\liminf_{x \rightarrow 0^+} f(x) = 0 \geq f(0)$

② Let $A = (0, 1)$ $f(x) = x \mathbb{1}_{[x \in (0, 1)]} + \infty \mathbb{1}_{[x \notin (0, 1)]}$ is NOT l.s.c.

Proposition. A function $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ is l.s.c. iff $\text{epi } f$ is closed.

Proof. " $\Rightarrow$ ": Assume that f is l.s.c. and $(x_n, t_n)_{n \geq 1} \in \text{epi } f \Rightarrow t_n \geq f(x_n)$

Also, $(x_n, t_n) \rightarrow (\bar{x}, \bar{t})$. $\bar{t} = \lim_{n \rightarrow \infty} t_n \geq \liminf_{n \rightarrow \infty} f(x_n) \geq f(\bar{x})$

Hence $(\bar{x}, \bar{t}) \in \text{epi } f$.

" $\Leftarrow$ ": Assume that $\text{epi } f$ is closed. If f is not l.s.c., $\exists (x_n, t_n)_{n \geq 1} \rightarrow (\bar{x}, \bar{t})$ s.t. $\bar{t} = \lim_{n \rightarrow \infty} f(x_n) < f(\bar{x})$ and $t_n = f(x_n)$. By definition, $(x_n, t_n) \in \text{epi } f$ and thus $(\bar{x}, \bar{t}) \in \text{epi } f$, which is a contradiction. ■

Lemma. Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be a convex function such that $\text{epi } f$ is convex.

Then, there exists $cl f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ s.t. $\text{epi } (cl f) = \overline{\text{epi } f}$ (and $cl f$ is l.s.c.).

Proof. Let $g(x) = \liminf_{x' \in \mathbb{R}^d, \|x-x'\| \leq r} f(x')$ is l.s.c.

Then, ① $g(x) \leq f(x) \quad \forall x \in \mathbb{R}^d$

② $g(x) = f(x) \quad \forall x \in \text{relint dom } f$

$\text{dom } f \subseteq \text{dom } g \subseteq \overline{\text{dom } g}$
 $\text{relint dom } f = \text{relint dom } g. \quad \left. \begin{array}{l} \text{dom } f \subseteq \text{dom } g \subseteq \overline{\text{dom } g} \\ \text{relint dom } f = \text{relint dom } g \end{array} \right\} \Rightarrow \text{dom } g = \overline{\text{dom } f}.$

Lecture 20

Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex. Then,

(i) there exists $cl f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ such that $\text{epi}(cl f) = \overline{\text{epi} f}$.

(ii) $cl f(x) = \lim_{x' \rightarrow x} f(x')$ and $\text{dom} f \subseteq \text{dom}(cl f) \subseteq \overline{\text{dom} f}$

and $\text{ri dom} f = \text{ri dom}(cl f)$

(iii) $cl f(x) = \sup \{ \phi(x) : \phi(y) = A^T y + b \leq f(y) \}$

Proof. (i) Let $cl f(x) := \inf \{ t : (x, t) \in \overline{\text{epi} f} \}$ with the convention that $\inf \emptyset = +\infty$.

$(x, t) \in \overline{\text{epi} f} \Rightarrow \begin{cases} (x_n, t_n) \in \text{epi} f \\ (x_n, t_n) \rightarrow (x, t) \end{cases} \Rightarrow (x_n, t_n + c) \in \text{epi} f \quad \forall c \geq 0.$

$\Rightarrow (x, t+c) = \lim_{n \rightarrow \infty} (x_n, t_n + c) \in \overline{\text{epi} f}$

$\Rightarrow \overline{\text{epi} f} = \text{epi}(cl f)$

(ii) ① Since $\overline{\text{epi} f} = \text{epi}(cl f)$ is closed. Then, $cl f$ is lower-semicontinuous and $cl f(x) \leq f(x)$,

$\forall x \in \mathbb{R}^n$, since $\text{epi}(cl f) \supseteq \text{epi} f$. Hence, $cl f(x) \leq \lim_{x' \rightarrow x} cl f(x') \leq \lim_{x' \rightarrow x} f(x')$

② $(x, cl f(x)) \in \text{epi}(cl f) = \overline{\text{epi} f}$
 $\Rightarrow \exists (x_n, t_n) \in \text{epi} f \Rightarrow f(x_n) \leq t_n$

$(x_n, t_n) \rightarrow (x, cl f(x))$

$\Rightarrow cl f(x) = \lim_{n \rightarrow \infty} t_n \geq \lim_{n \rightarrow \infty} f(x_n)$

$\geq \lim_{x' \rightarrow x} f(x')$

$\Rightarrow cl f(x) = \lim_{x' \rightarrow x} f(x').$

(iii) ① $\forall x \in \text{dom} f$, $f(x) < +\infty$, i.e. $cl f(x) \leq f(x) < +\infty \Rightarrow x \in \text{dom}(cl f)$

$\exists (x_n, f(x_n)) \rightarrow (x, cl f(x))$ and $x_n \in \text{dom} f \Rightarrow x \in \overline{\text{dom} f}$.

③ If $x \in \text{int dom}(cl f)$, then $cl f(y) < +\infty \quad \forall y \in B_\epsilon(x)$

Hence, $\text{dom} f$ is dense in $B_\epsilon(x)$.

Since $\text{dom} f$ is convex, $\text{int } B_\epsilon(x) \subseteq \text{dom} f \Rightarrow x \in \text{int}(\text{dom} f)$

③ $g(x) := \sup \{ \phi(x) : \phi(y) = A^T y + b \leq f(y) \quad \forall y \in \mathbb{R}^n \}$.

Then, $\forall \phi$ where ϕ is continuous, $g(\cdot)$ is lower-semicontinuous as supremum of continuous functions.

$\forall \phi$ where ϕ is convex, $g(\cdot)$ is convex.

(a) $\forall x \in \text{ri dom} f$, $g(x) \leq f(x)$ and $\exists v \in \partial f(x)$ s.t. $f(y) \geq f(x) + v^T(y-x), \forall y \in \mathbb{R}^n$

$= v^T y + (f(x) - v^T x)$

$= \phi(y).$

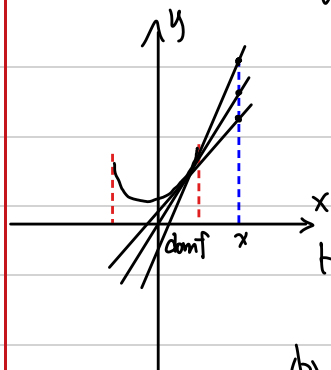
and $f(x) = \phi(x)$

Hence, $g(x) = \sup \{ \phi(x) : \dots \} \geq f(x)$

$f(x) = g(x) = cl f(x).$

(b) $\forall x \notin \overline{\text{dom} f}$, $f(y) = +\infty, cl f(y) = +\infty, \forall y \in B_\epsilon(x).$

$\forall t, \exists \phi(y) = A^T y + b \leq f(y)$ and $\phi(x) \geq t \Rightarrow g(x) = \sup \{ \phi(x) : \dots \} = +\infty$



$\left. \begin{array}{l} cl f \text{ is convex and l.s.c.} \\ g \text{ is convex, l.s.c.} \\ cl f = g \text{ on ri } A \end{array} \right\} \Rightarrow cl f = g$
 (on the relative boundary of A)

Definition (Conjugate Function) Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$, the conjugate of f is defined as

$$f^*(d) := \sup_{x \in \mathbb{R}^n} (d^T x - f(x)) \quad \forall d \in \mathbb{R}^n$$

$$= \sup \{ \phi_x(d) : \phi_x(v) = v^T x - f(x), \quad \forall v \in \mathbb{R}^n \}$$

Remark. ① $v \mapsto \phi_x(v)$ is affine (continuous and convex) $\Rightarrow f^*(d)$ is lower-semicontinuous and convex.

$$② 0 \in \text{dom}(f^*) \Leftrightarrow f^*(0) < +\infty \Leftrightarrow \sup_{x \in \mathbb{R}^n} (-f(x)) < +\infty \Leftrightarrow \inf_{x \in \mathbb{R}^n} f(x) > -\infty$$

Proposition. Let $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ be convex. Then, the bi-conjugate function of f

$$(f^*)^*(x) := \sup_{d \in \mathbb{R}^n} (x^T d - f^*(d)) = \text{cl } f.$$

Proof. $(f^*)^*(x) = \sup_{d \in \mathbb{R}^n} (x^T d - f^*(d))$

$$= \sup_{d \in \mathbb{R}^n} (x^T d - a)$$

Since $a \geq f^*(d)$, $a \geq \sup_{y \in \mathbb{R}^n} (d^T y - f(y)) \Leftrightarrow a \geq d^T y - f(y) \quad \forall y \in \mathbb{R}^n \Leftrightarrow f(y) \geq d^T y - a \quad \forall y \in \mathbb{R}^n$.

we have $(f^*)^*(x) = \sup_{d \in \mathbb{R}^n} \{ \underbrace{x^T d - a}_{\phi(x)} : \underbrace{f(y) \geq d^T y - a}_{\phi(y)} \quad \forall y \in \mathbb{R}^n \}$

$$= \sup \{ \phi(x) : \phi(y) \text{ is affine and } \phi(y) \leq f(y), \quad \forall y \in \mathbb{R}^n \} = \text{cl } f$$

Remark. ① For any $f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\}$ not necessary convex. We have

$$(f^*)^* = \sup \{ \phi(\cdot) : \phi(\cdot) \text{ is affine, } \phi(y) \leq f(y), \quad \forall y \}$$

$$= \sup \{ \phi(\cdot) : \phi(\cdot) \text{ is convex, } \phi(y) \leq f(y), \quad \forall y \}$$

is called the convex envelope of $f(\cdot)$.

$$② 0 \in \text{dom}(f^*) \Leftrightarrow f^*(0) < +\infty \Leftrightarrow \sup_{x \in \mathbb{R}^n} (-f(x)) < +\infty \Leftrightarrow \inf_{x \in \mathbb{R}^n} f(x) > -\infty$$

$$③ \text{ Let } f: \mathbb{R}^n \rightarrow \mathbb{R} \cup \{+\infty\} \text{ be proper, } f^*(d) \in \mathbb{R} \cup \{+\infty\}.$$

$$④ \text{ Furthermore, } \hat{x} \in \partial f(0) \text{ iff } f^*(d) \geq f^*(0) + x^T d \quad \forall d \in \mathbb{R}^n.$$

$$f(\hat{x}) = (f^*)^*(\hat{x}) = \sup_{d \in \mathbb{R}^n} (\hat{x}^T d - f^*(d))$$

$$\leq \sup_{d \in \mathbb{R}^n} (\hat{x}^T d - (f^*(0) + \hat{x}^T d)) = f^*(0) = -\sup_{x \in \mathbb{R}^n} (-f(x)) = \inf_{x \in \mathbb{R}^n} f(x).$$

$\hat{x}$ is a minimizer of f , where f is convex.

Example. ① $f(x) \equiv -a$. $f^*(d) = \sup_x (d^T x - f(x))$

$$= \sup_x (d^T x + a)$$

$$= \begin{cases} a, & d=0 \\ +\infty, & d \neq 0 \end{cases}$$

$$② f(x) = A^T x + b. \quad f^*(d) = \sup_x (d^T x - A^T x - b)$$

$$= \begin{cases} -b, & d=0 \\ +\infty, & d \neq 0 \end{cases}$$

$$③ f(x) = \frac{1}{2} x^T A x. \quad f^*(d) = \sup_x (d^T x - \frac{1}{2} x^T A x)$$

$$= \frac{1}{2} d^T A^{-1} d$$

$$④ L(x, \lambda) = f(x) + \lambda g(x)$$

$$d(\lambda) := \sup_{x \in \mathbb{R}^n} L(x, \lambda)$$

When $g(x) = x$, $d(\lambda) = \sup_{x \in \mathbb{R}^n} (\lambda x - f(x))$

$$= (-f^*)(\lambda)$$

Lecture 21 IV. Numerical Method

Consider the optimization problem

$$\inf_{x \in K} f(x)$$

§ Projected Gradient Descent

Algorithm: $x^{(k+1)} \leftarrow \text{Proj}_K(x^{(k)} - \eta \nabla f(x^{(k)}))$

- Definition/Uniqueness of $\text{Proj}_K y$ (§ Lecture 8)

① The map $x \mapsto \|x - y\|^2$ is strictly convex and coercive.

② K is closed and hence, a minimizer exists. It follows that such minimizer x^* is unique by the strict convexity.

③ $\langle y - x^*, z - x^* \rangle \leq 0 \quad \forall z \in K$

$\forall z \in K, t \in (0, 1], tz + (1-t)x^* \in K$, we have $\|tz + (1-t)x^* - y\|^2 \geq \|x^* - y\|^2$

$$\Rightarrow \|t(z - x^*) + x^* - y\|^2 \geq \|x^* - y\|^2$$

$$t^2 \|z - x^*\|^2 + 2t \langle z - x^*, x^* - y \rangle + \|x^* - y\|^2$$

$$\Rightarrow 2t \langle z - x^*, x^* - y \rangle + t^2 \|z - x^*\|^2 \geq 0$$

$$\Rightarrow \langle y - x^*, z - x^* \rangle \leq \frac{t}{2} \|z - x^*\|^2 \quad \forall t \in (0, 1]$$

$$\Rightarrow \langle y - x^*, z - x^* \rangle \leq 0$$

④ Conversely, $\langle y - x^*, z - x^* \rangle \leq 0 \quad \forall z \in K$ implies that

$$\begin{aligned} \|y - z\|^2 - \|y - \hat{x}\|^2 &= \|y - \hat{x} + \hat{x} - z\|^2 - \|y - \hat{x}\|^2 \\ &= \|y - \hat{x}\|^2 + 2 \langle y - \hat{x}, \hat{x} - z \rangle + \|\hat{x} - z\|^2 - \|y - \hat{x}\|^2 \\ &\geq 0 \end{aligned}$$

$\forall z \in K, \|y - z\| \geq \|y - \hat{x}\|, \hat{x} = \text{Proj}_K(y)$.

④ $\|\text{Proj}_K(y_1) - \text{Proj}_K(y_2)\| \leq \|y_1 - y_2\|$

Proof. $\langle y_1 - \text{Proj}_K(y_1), \text{Proj}_K(y_2) - \text{Proj}_K(y_1) \rangle \leq 0, \langle y_2 - \text{Proj}_K(y_2), \text{Proj}_K(y_1) - \text{Proj}_K(y_2) \rangle \leq 0$

Hence, $\langle y_1 - y_2 - (\text{Proj}_K(y_1) - \text{Proj}_K(y_2)), \text{Proj}_K(y_2) - \text{Proj}_K(y_1) \rangle \leq 0$

$$\|\text{Proj}_K(y_2) - \text{Proj}_K(y_1)\|^2 \leq \langle y_1 - y_2, \text{Proj}_K(y_1) - \text{Proj}_K(y_2) \rangle$$

$$\leq \|y_1 - y_2\| \cdot \|\text{Proj}_K(y_1) - \text{Proj}_K(y_2)\|$$

Lecture 22

Theorem. Assume that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is elliptic in the sense that $H_f(x) - \alpha I$ is PSD for some $\alpha > 0$ and $\nabla f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is M -Lipschitz and $\rho \in (0, \frac{2\alpha}{M^2})$. Then, $x^{(k)} \rightarrow x^*$, where x^* is the minimizer of $\min_{x \in K} f(x)$.

Proof. $f(x) - f(0) = \langle \nabla f(0), x \rangle + \frac{1}{2} x^T H_f(\xi) x$ for some ξ
 $\geq \frac{1}{2} \alpha \|x\|^2$

Hence, f is coercive, i.e. $\lim_{x \rightarrow \infty} f(x) = +\infty$

Since K is closed, x^* exists.

K is convex
 f is strictly convex $\} \Rightarrow x^*$ is unique.

Lemma. ① $\langle \nabla f(y) - \nabla f(x^*), y - x^* \rangle \geq \alpha \|y - x^*\|^2, \forall y \in K$

② $\text{Proj}_K(f(x^*) - \rho \nabla f(x^*)) = x^*$

Proof. ① Let $\phi(t) := \langle \nabla f(ty + (1-t)x^*), y - x^* \rangle$

$$\Rightarrow \langle \nabla f(y) - \nabla f(x^*), y - x^* \rangle = \phi(1) - \phi(0) = \int_0^1 \phi'(t) dt$$

$$\text{Moreover, } \phi'(t) = \langle H_f(ty + (1-t)x^*)(y - x^*), (y - x^*) \rangle \geq \alpha \|y - x^*\|^2$$

$$\Rightarrow \langle \nabla f(y) - \nabla f(x^*), y - x^* \rangle = \int_0^1 \phi'(t) dt \geq \alpha \|y - x^*\|^2$$

② $\forall z \in K, f(tz + (1-t)x^*) - f(x^*) \geq 0$

$$\text{Let } t \in (0, 1), \text{ we have } \lim_{t \rightarrow 0} \frac{f(tz + (1-t)x^*) - f(x^*)}{t} \geq 0$$

$$\Rightarrow \langle \nabla f(x^*), z - x^* \rangle \geq 0, \forall z \in K$$

$$\Rightarrow \langle x^* - \rho \nabla f(x^*) - x^*, z - x^* \rangle = -\rho \langle \nabla f(x^*), z - x^* \rangle \leq 0, \forall z \in K$$

Recall that $x^* \in K$ and $\langle y - x^*, z - x^* \rangle \leq 0 \forall z \in K \Rightarrow x^* = \text{Proj}_K(y)$

we have $x^* = \text{Proj}_K(x^* - \rho \nabla f(x^*))$ ■

Convergence to a minimizer:

$$\text{We consider } \|x^{(k+1)} - x^*\| = \|\text{Proj}_K(x^{(k)} - \rho \nabla f(x^{(k)})) - x^*\|^2$$

$$= \|\text{Proj}_K(x^{(k)} - \rho \nabla f(x^{(k)})) - \text{Proj}_K(x^* - \rho \nabla f(x^*))\|^2$$

$$\leq \|x^{(k)} - \rho \nabla f(x^{(k)}) - (x^* - \rho \nabla f(x^*))\|^2$$

$$= \|x^{(k)} - x^*\|^2 + 2\rho \langle x^{(k)} - x^*, \nabla f(x^{(k)}) - \nabla f(x^*) \rangle + \rho^2 \|\nabla f(x^{(k)}) - \nabla f(x^*)\|^2$$

$$\leq (1 - 2\rho\alpha + M^2\rho^2) \|x^{(k)} - x^*\|^2$$

$$\text{As } \rho \in (0, \frac{2\alpha}{M^2}), 1 - 2\rho\alpha + M^2\rho^2 < 1 \text{ and } \|x^{(k)} - x^*\| \leq (1 - 2\rho\alpha + M^2\rho^2)^k \|x^{(0)} - x^*\|$$
 ■

Convergence rate: size of k w.r.t. $\frac{1}{\varepsilon}$.

§ Uzawa Algorithm

Consider the problem $\min_{Ax=b} f(x)$ where $f(x)$ is convex. $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.
($m \leq n$)

Remark. $L(x, \mu) = f(x) + \mu^T(Ax - b)$, $\mu \in \mathbb{R}^m$
 $d(\mu) := \inf_{x \in \mathbb{R}^n} L(x, \mu)$. $D := \sup_{\mu \in \mathbb{R}^m} d(\mu)$

Theorem. Assume that x^* exists. Then, $P = D$. and $\exists \mu^*$ st. $d(\mu^*) = P = D = L(x^*, \mu^*)$

Algorithm.

$$\begin{aligned} \mu^{(k+1)} &\leftarrow \mu^{(k)} + \rho \nabla d(\mu^{(k)}) \\ x^{(k+1)} &\leftarrow \arg\min_{x \in \mathbb{R}^n} f(x) + \langle \mu^{(k)}, Ax - b \rangle \end{aligned}$$

Theorem. $\lambda \mapsto d(\lambda)$ is concave. If $x_0 \in \mathbb{R}^n$ satisfies $L(x_0, \mu_0) = d(\mu_0) = \inf_{x \in \mathbb{R}^n} L(x, \mu_0)$
 Then, $\nabla d(\lambda_0) = Ax_0 - b$.

Proof. ① $\lambda \mapsto L(x, \lambda) = f(x) + \langle \lambda, Ax - b \rangle$ is affine.

Hence, $d(\lambda)$ is the infimum of affine functions

$$\begin{aligned} \text{② } \forall v \in \mathbb{R}^n, h \in \mathbb{R}, d(\mu_0 + hv) &\leq L(x_0, \mu_0 + hv) = f(x_0) + \langle \mu_0 + hv, Ax_0 - b \rangle \\ &= f(x_0) + \langle \mu_0, Ax_0 - b \rangle + h \langle v, Ax_0 - b \rangle \\ &= d(\mu_0) + h \langle v, Ax_0 - b \rangle \end{aligned}$$

$$\Rightarrow \frac{d(\mu_0 + hv) - d(\mu_0)}{h} \geq \langle v, Ax_0 - b \rangle$$

$$\Rightarrow \langle \nabla d(\lambda_0), v \rangle \geq \langle v, Ax_0 - b \rangle \quad \forall v$$

Replacing v by $-v$, we obtain $\langle \nabla d(\lambda_0), -v \rangle \geq \langle -v, Ax_0 - b \rangle$
 $\Rightarrow \nabla d(\lambda_0) = Ax_0 - b$

Theorem. Assume that $f: \mathbb{R}^n \rightarrow \mathbb{R}$ s.t. $Hf(x) - \alpha I$ is PSD for some $\alpha > 0$
 and $\|A\|^2 = \max_{\|x\|=1} x^T A A^T x$ and $\rho \in (0, \frac{2\alpha}{\|A\|^2})$. Then, $x^{(k)} \rightarrow x^*$. Moreover, if $\text{rank } A = m$, then $\mu^{(k)} \rightarrow \mu^*$.

Proof. $\|\mu^{(k+1)} - \mu^*\|^2 = \|\mu^{(k)} + \rho(Ax^{(k)} - b) - \mu^*\|^2 = \|\mu^{(k)} + \rho(Ax^{(k)} - b) - (\mu^* + \rho(Ax^* - b))\|^2$
 Note that $\mu^* + \rho(Ax^* - b) = \mu^*$.
 $\nabla L(x^{(k)}, \mu^{(k)}) = \nabla f(x^{(k)}) + A^T \mu^{(k)} = 0$
 $\Rightarrow A^T(\mu^{(k)} - \mu^*) = -\nabla f(x^{(k)}) + \nabla f(x^*)$
 $\Rightarrow \|\mu^{(k+1)} - \mu^*\| \leq \|\mu^{(k)} - \mu^*\|^2 + 2\rho \langle \nabla f(x^{(k)}) - \nabla f(x^*), x^{(k)} - x^* \rangle + \rho^2 \|A(x^{(k)} - x^*)\|^2$
 $\leq \|\mu^{(k)} - \mu^*\|^2 - \rho(2\alpha - \rho\|A\|^2) \|x^{(k)} - x^*\|^2$

$$\text{① } \|x^{(k)} - x^*\|^2 \leq \frac{1}{\rho(2\alpha - \rho\|A\|^2)} (\|\mu^{(k)} - \mu^*\|^2 - \|\mu^{(k+1)} - \mu^*\|^2)$$

$$\text{② } \|\mu^{(k+1)} - \mu^*\| \leq \|\mu^{(k)} - \mu^*\|^2 \searrow \pi \geq 0$$

$$\Rightarrow \|\mu^{(k+1)} - \mu^*\| - \|\mu^{(k)} - \mu^*\| \rightarrow 0 \Rightarrow \|x^{(k)} - x^*\|^2 \rightarrow 0$$

$\nabla f(x^{(k)}) + A^T \mu^{(k)} = 0$ The sequence of $\mu^{(k)}$ is uniformly bounded. Take a subsequence $\mu^{(k_i)} \rightarrow \hat{\mu}$.

$$\nabla f(x^*) + A^T \hat{\mu} = 0, \nabla f(x^*) + A^T \mu^* = 0 \Rightarrow \hat{\mu} = \mu^*$$

Lecture 23

We are interested in solving the convex optimization problem:

$$(P): \inf f(x) \quad \text{subject to } g_1(x) \leq 0, g_2(x) \leq 0, \dots, g_m(x) \leq 0$$

Conditioned on:

- ① $f(\cdot), g_1(\cdot), g_2(\cdot), \dots, g_m(\cdot)$ are all convex

$$\textcircled{2} \text{ Hess}(f) \succeq \alpha I_n$$

$$\textcircled{3} g_1(\cdot), g_2(\cdot), \dots, g_m(\cdot) \text{ are } L\text{-Lipschitz}$$

$$\textcircled{4} \eta \in (0, \frac{2\alpha}{L^2}).$$

$$D = \max_{\lambda \in \mathbb{R}_+^m} d(\lambda) \quad \text{where } d(\lambda) := \inf_{x \in \mathbb{R}^n} f(x) + \sum_{i=1}^m \lambda_i g_i(x)$$

Recall that: Under Slater's condition, $P = D$, and $\exists \lambda^* \in \mathbb{R}_+^m$ st. $d(\lambda^*) = D = f(x^*) + \sum \lambda_i^* g_i(x^*)$

Algorithm.

$$\begin{aligned} \lambda^{(k+1)} &\leftarrow \text{Proj}_{\mathbb{R}_+^m} (\lambda^{(k)} + \eta \nabla d(\lambda^*)) = \text{Proj}_{\mathbb{R}_+^m} (\lambda^{(k)} + \eta g(x^{(k)})) \\ x^{(k+1)} &\leftarrow \arg\min_{x \in \mathbb{R}^n} (f(x) + \sum_{i=1}^m \lambda_i^{(k)} g_i(x)) \end{aligned}$$

Theorem. Under the given conditions, there exists a unique optimal solution to (P) and $x^{(k)} \rightarrow x^*$ as $k \rightarrow \infty$.

Proof. ① Existence of x^* is guaranteed by the coercivity of f and closedness of K . (Minimization of a continuous, coercive function on a closed set).

$$\textcircled{2} d(\lambda^*) = \min_{\lambda \in \mathbb{R}_+^m} d(\lambda)$$

$$\text{By KKT theorem, } 0 = \sum_{i=1}^m \lambda_i^* g_i(x^*) \geq \sum_{i=1}^m \lambda_i g_i(x^*)$$

$$\Rightarrow \sum_{i=1}^m (\lambda_i^* - \lambda_i) g_i(x^*) \geq 0 \Leftrightarrow \langle \lambda^* - \lambda, g(x^*) \rangle \geq 0$$

$$\Rightarrow \langle \lambda^* - \lambda, \lambda^* - (\lambda^* + \eta g(x^*)) \rangle \leq 0 \quad \forall \lambda \in \mathbb{R}_+^m$$

$$\Rightarrow \lambda^* = \text{Proj}_{\mathbb{R}_+^m} (\lambda^* + \eta g(x^*))$$

$$\begin{aligned} \textcircled{3} \|\lambda^{(k+1)} - \lambda^*\|^2 &= \|\text{Proj}_{\mathbb{R}_+^m} (\lambda^{(k)} - \eta g(x^{(k)})) - \text{Proj}_{\mathbb{R}_+^m} (\lambda^* + \eta g(x^*))\|^2 \\ &\leq \|\lambda^{(k)} - \eta g(x^{(k)}) - (\lambda^* + \eta g(x^*))\|^2 \\ &\leq \|\lambda^{(k)} - \lambda^*\|^2 + 2\eta \langle \lambda^{(k)} - \lambda^*, g(x^{(k)}) - g(x^*) \rangle + \eta^2 \|g(x^{(k)}) - g(x^*)\|^2 \end{aligned}$$

$$\text{By Euler's First Order Conditions: } \nabla f(x^*) + \langle \lambda^*, \nabla g(x^*) \rangle = 0$$

$$\nabla f(x^{(k)}) + \langle \lambda^{(k)}, \nabla g(x^{(k)}) \rangle = 0$$

$$\Rightarrow \nabla f(x^*) - \nabla f(x^{(k)}) + \langle \lambda^*, \nabla g(x^*) \rangle - \langle \lambda^{(k)}, \nabla g(x^{(k)}) \rangle = 0$$

$$\text{Besides, } \langle \nabla f(x^*) - \nabla f(x^{(k)}), x^* - x^{(k)} \rangle \geq \alpha \|x^* - x^{(k)}\|^2 \text{ since } \text{Hess}(f) \succeq \alpha I_n$$

$$\sum_{i=1}^m \lambda_i^* \langle \nabla g_i(x^*), x^{(k)} - x^* \rangle + \sum_{i=1}^m \lambda_i^{(k)} \langle \nabla g_i(x^{(k)}), x^* - x^{(k)} \rangle$$

$$\geq \alpha \|x^* - x^{(k)}\|^2 \leq \underbrace{g_i(x^{(k)}) - g_i(x^*)}_{\leq g_i(x^*) - g_i(x^{(k)})} \quad \text{by convexity}$$

$$\Rightarrow \sum_{i=1}^m (\lambda_i^* - \lambda_i^{(k)}) (g_i(x^{(k)}) - g_i(x^*)) \geq \alpha \|x^* - x^{(k)}\|^2$$

$$\Rightarrow \langle \lambda^* - \lambda^{(k)}, g(x^*) - g(x^{(k)}) \rangle \geq \alpha \|x^* - x^{(k)}\|^2$$

$$\text{Hence, } \|\lambda^{(k+1)} - \lambda^*\|^2 \leq \|\lambda^{(k)} - \lambda^*\|^2 - 2\eta \alpha \|x^{(k)} - x^*\|^2 + \eta^2 L^2 \|x^{(k)} - x^*\|^2$$

$$\Rightarrow \|x^{(k)} - x^*\|^2 \geq \frac{\|\lambda^{(k+1)} - \lambda^*\|^2 - \|\lambda^{(k)} - \lambda^*\|^2}{\eta^2 L^2 - 2\eta \alpha} \quad \begin{matrix} k \rightarrow \|\lambda^{(k)} - \lambda^*\| \text{ is decreasing, hence converges} \\ \end{matrix}$$

Lecture 24 - Application: Markowitz Portfolio Selection

Background: $X = (x_1, x_2, \dots, x_n)$ random vector

$$\text{Let } m = E[X] = (E[x_1], E[x_2], \dots, E[x_n])$$

$$\Sigma = \text{Var}[X] = \begin{pmatrix} \text{Var}[X_1] & \text{Cov}(X_1, X_2) & \dots & \text{Cov}(X_1, X_n) \\ \text{Cov}(X_2, X_1) & \text{Var}[X_2] & \dots & \text{Cov}(X_2, X_n) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(X_n, X_1) & \text{Cov}(X_n, X_2) & \dots & \text{Var}[X_n] \end{pmatrix}$$

$$\text{Then, } E[b^T X] = b^T m \text{ and } \text{Var}[b^T X] = b^T \Sigma b \text{ for any fixed } b \in \mathbb{R}^n.$$

Consider the stock setting:

Stock price S_t^k $t=0$ (buy), $t=1$ (sell), $k=1, 2, \dots, n$ (# number of stocks)

$$\text{Return } R^k := S_1^k - S_0^k \quad (S_1^k \text{ is R.V.})$$

$$\text{Relative return } X^k := \frac{R^k}{S_0^k}, \text{ return per \$ of investment}$$

$$\text{Portfolio } w = (w_1, w_2, \dots, w_n), \sum_{k=1}^n w_k = 1$$

$$\text{Return of the portfolio: } R := \sum_{k=1}^n w_k X_k$$

$$\text{Formalization: } \max E[R] / \min \text{Var}[R]$$

① Fix a σ_0 we accept all portfolio where the risk, $\text{Var}[R] \leq \sigma_0^2$. $\rightarrow$ dual

② Fix a v_0 we accept all portfolio where the return $E[R] \geq v_0$.

$$(P) \min_w w^T \Sigma w \text{ subject to } w^T m \geq v_0, w^T \mathbf{1} = 1 \quad m \in \mathbb{R}^n, \Sigma \in S^n, v_0 \in \mathbb{R} \text{ are given}$$

Assumption: $\Sigma \in S_{++}^n$, $n \geq 2$, $\exists w$ s.t. $m^T w > \mu$. $\left. \begin{matrix} \text{① Coercive} \\ \text{② Closed set} \end{matrix} \right\}$ Admits a minimizer

$$\text{Solution: } \min_{\substack{w^T m \geq v \\ w^T \mathbf{1} = 1}} f(w) \Rightarrow -f(w) = \frac{1}{2} w^T \Sigma w, g(w) = \mu - m^T w, h(w) = \mathbf{1}^T w - 1.$$

Qualification: Slater's condition (affinity) and $\exists w$ s.t. $g(w) < 0, \nabla h(w) = \mathbf{1}$ linearly independent

$$\nabla f(w) = \Sigma w, \nabla g(w) = -m, \nabla h(w) = \mathbf{1}$$

$$\text{KKT: } \exists \lambda \in \mathbb{R}_+, \gamma \in \mathbb{R} \text{ s.t. } \begin{cases} \nabla f(w^*) + \lambda \nabla g(w^*) + \gamma \nabla h(w^*) = 0 \\ \lambda g(w^*) = 0 \end{cases}$$

$$\begin{cases} \Sigma w^* - \lambda m + \gamma \mathbf{1} = 0 \\ \lambda (\mu - m^T w^*) = 0 \end{cases}$$

$$\text{Case 1. } \mu - m^T w > 0 \Rightarrow \lambda = 0$$

$$\Rightarrow w^* = -\Sigma^{-1} \gamma \mathbf{1} \quad \text{By } \mathbf{1}^T w^* = 1 \text{ we have } w^* = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}} := w_{mv}^*$$

$$\text{Case 2. } \mu - m^T w^* = 0 \Rightarrow \lambda > 0$$

$$\begin{cases} \mathbf{1}^T w^* = 1 \\ \Sigma w^* - \lambda m + \gamma \mathbf{1} = 0 \end{cases}$$

① Det $M = 0$ if $m = c\mathbf{1}$

② Det $M > 0$ otherwise

$$w^* = \Sigma^{-1} (\Sigma w^*) = \lambda \Sigma^{-1} m - \gamma \Sigma^{-1} \mathbf{1}$$

$$\begin{cases} m^T (\lambda \Sigma^{-1} m - \gamma \Sigma^{-1} \mathbf{1}) = \mu \\ \mathbf{1}^T (\lambda \Sigma^{-1} m - \gamma \Sigma^{-1} \mathbf{1}) = 1 \end{cases} \Leftrightarrow \begin{pmatrix} m^T \Sigma^{-1} m & m^T \Sigma^{-1} \mathbf{1} \\ \mathbf{1}^T \Sigma^{-1} m & \mathbf{1}^T \Sigma^{-1} \mathbf{1} \end{pmatrix} \begin{pmatrix} \lambda \\ -\gamma \end{pmatrix} = \begin{pmatrix} \mu \\ 1 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} \lambda \\ -\gamma \end{pmatrix} = \begin{pmatrix} \mathbf{1}^T v \\ m^T v \end{pmatrix}, v := \Sigma^{-1} (\mu \mathbf{1} - m) \Rightarrow w^* = (1-\alpha) \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}} + \alpha \frac{\Sigma^{-1} m}{\mathbf{1}^T \Sigma^{-1} m}$$

$$\Rightarrow w^* = (1-\alpha) w_{mv}^* + \alpha w_{mk}^*$$

$$\alpha = \frac{\mu (m^T \Sigma^{-1} \mathbf{1}) (\mathbf{1}^T \Sigma^{-1} \mathbf{1}) - (m^T \Sigma^{-1} \mathbf{1})^2}{\delta}$$

Corollary 1 (Frank-Wolfe Theorem). Suppose that S is the polyhedron

$$S := \{x \in \mathbb{R}^n \mid Ax \leq b; Ex = d\}, \text{ where } A \in \mathbb{R}^{m \times n}, E \in \mathbb{R}^{l \times n}, b \in \mathbb{R}^m \text{ and } d \in \mathbb{R}^l$$

whose recession cone is $\text{rec } S := \{p \in \mathbb{R}^n \mid Ap \leq 0_m; Ep = 0_l\}$.

We also suppose that

$$f(x) := \frac{1}{2} x^T Q x + q^T x, \text{ where } x \in \mathbb{R}^n, Q \in \mathbb{R}^{n \times n} \text{ is a PSD matrix, and } q \in \mathbb{R}^n.$$

which has a recession cone $\text{rec } f = \{p \in \mathbb{R}^n \mid Qp = 0_n; q^T p \leq 0\}$.

Then, the problem $\min_{x \in S} f(x)$ is a convex quadratic problem and the following three statements are equivalent:

(a) The optimization problem has a non-empty (closed and convex) set of globally optimal solutions.

(b) f is lower bounded on S

(c) For every vector p in the intersection of the recession cone $\text{rec } S$ to S and the null space $\mathcal{N}(Q)$ of the matrix Q , it holds that $q^T p \geq 0$.

In other words,

$$p \in \text{rec } S \cap \mathcal{N}(Q) \Rightarrow q^T p \geq 0.$$

Improvement to the Weierstrass Theorem: If f is convex on S , and S is non-empty, closed and convex, then the problem has a non-empty, convex and compact set of globally optimal solutions if and only if $\text{rec } S \cap \text{rec } f = \{0_n\}$.

Example. Consider a special case of $f(x) := \frac{1}{x}$, $S := [1, +\infty)$

Then, f is bounded from below over S , but never attains the infimum on S . Therefore, the problem has no optimal solution. It is clear that f is not a polynomial function.

Corollary. For the general linear programming problem, the following three statements are identical:

(a) The optimization problem has a non-empty (closed and convex) set of globally optimal solutions.

(b) f is lower bounded on S

(c) $p \in \text{rec } S \Rightarrow q^T p \geq 0$